Divide and Conquer

1

Question 1

CORRECT

Which of the following algorithms is NOT a divide & conquer algorithm by nature?

A Euclidean algorithm to compute the greatest common divisor

Heap Sort

C Cooley-Tukey fast Fourier transform

D Quick Sort

Divide and Conquer
Discuss it

Question 1 Explanation:
See Divide and Conquer

Question 2

CORRECT

Consider the following C program

```
int main()
{
   int x, y, m, n;
   scanf ("%d %d", &x, &y);
   /* x > 0 and y > 0 */
   m = x; n = y;
   while (m != n)
   {
      if(m>n)
         m = m - n;
      else
         n = n - m;
   }
   printf("%d", n);
}
```

Run on IDE

What does the program compute? (GATE CS 2004)

A x + y using repeated subtraction

B x mod y using repeated subtraction

the greatest common divisor of x and y

D the least common multiple of x and y

Divide and Conquer
Discuss it

Question 2 Explanation:
This is an implementation of Euclid's algorithm to find GCD

Question 3

CORRECT

Consider the polynomial p(x) = a0 + a1x + a2x^2 +a3x^3, where ai != 0, for all i. The minimum number of multiplications needed to evaluate p on an input x is:

3

B 4

C 6

D 9

Divide and Conquer
Discuss it

Question 3 Explanation:
Multiplications can be minimized using following order for evaluation of the given expression. p(x) = a0 + x(a1 + x(a2 + a3x))

Question 4

CORRECT

Maximum Subarray Sum problem is to find the subarray with maximum sum. For example, given an array {12, -13, -5, 25, -20, 30, 10}, the maximum subarray sum is 45. The naive solution for this problem is to calculate sum of all subarrays starting with every element and return the maximum of all. We can solve this using Divide and Conquer, what will be the worst case time complexity using Divide and Conquer.

A O(n)

O(nLogn)

C O(Logn)

D O(n^2)

Divide and Conquer
Discuss it

Question 4 Explanation:
See http://www.geeksforgeeks.org/divide-and-conquer-maximum-sum-subarray/

Question 5

CORRECT

Consider a situation where you don't have function to calculate power (pow() function in C) and you need to calculate x^n where x can be any number and n is a positive integer. What can be the best possible time complexity of your power function?

A O(n)

B O(nLogn)

C O(LogLogn)

O(Logn)

Divide and Conquer
Discuss it

Question 5 Explanation:
We can calculate power using divide and conquer in O(Logn) time.
See http://www.geeksforgeeks.org/write-a-c-program-to-calculate-powxn/.

Question 6

WRONG

Consider the problem of searching an element x in an array 'arr[]' of size n. The problem can be solved in O(Logn) time if. 1) Array is sorted 2) Array is sorted and rotated by k. k is given to you and k <= n 3) Array is sorted and rotated by k. k is NOT given to you and k <= n 4) Array is not sorted

1 Only

B 1 & 2 only

1, 2 and 3 only

D 1, 2, 3 and 4

Divide and Conquer
Discuss it

Question 6 Explanation:
See http://www.geeksforgeeks.org/search-an-element-in-a-sorted-and-pivoted-array/

Question 7

CORRECT

The secant method is used to find the root of an equation f(x) = 0. It is started from two distinct estimates x_a and x_b for the root. It is an iterative procedure involving linear interpolation to a root. The iteration stops if $f(x_b)$ is very small and then x_b is the solution. The procedure is given below. Observe that there is an expression which is missing and is marked by? Which is the suitable expression that is to be put in place of? So that it follows all steps of the secant method? **Secant**

```
Initialize: xa, xb, ε, N     // ε = convergence indicator
fb = f(xb) i = 0
while (i < N and |fb| > ε) do
   i = i + 1                  // update counter
   xt = ?                     // missing expression for
                              // intermediate value
   xa = xb                    // reset xa
   xb = xt                    // reset xb
   fb = f(xb)                 // function value at new xb
end while
if |fb| > ε
  then // loop is terminated with i = N
  write "Non-convergence"
else
  write "return xb"
end if
```

A $x_b - (f_b - f(x_a))\, f_b / (x_b - x_a)$

B $x_a - (f_a - f(x_a))\, f_a / (x_b - x_a)$

C $x_b - (f_b - x_a)\, f_b / (x_b - f_b(x_a)$

$x_a - (x_b - x_a)\, f_a / (f_b - f(x_a))$

Divide and Conquer GATE-CS-2015 (Set 2)
Discuss it

Question 7 Explanation:
ttp://en.wikipedia.org/wiki/Secant_method

Question 8

CORRECT

Suppose you are provided with the following function declaration in the C programming language.

```
int partition (int a[], int n);
```

The function treats the first element of a[] as a pivot, and rearranges the array so that all elements less than or equal to the pivot is in the left part of the array, and all elements greater than the pivot is in the right part. In addition, it moves the pivot so that the pivot is the last element of the left part. The return value is the number of elements in the left part. The following partially given function in the C programming language is used to find the kth smallest element in an array a[ ] of size n using the partition function. We assume k ≤ n

```
int kth_smallest (int a[], int n, int k)
{
   int left_end = partition (a, n);
   if (left_end+1==k)
   {
       return a [left_end];
   }
   if (left_end+1 > k)
   {
      return kth_smallest (_____________________);
   }
   else
   {
      return kth_smallest (_____________________);
    }
}
```

Run on IDE

The missing argument lists are respectively

(a, left_end, k) and (a+left_end+1, n–left_end–1, k–left_end–1)

B (a, left_end, k) and (a, n–left_end–1, k–left_end–1)

C (a, left_end+1, N–left_end–1, K–left_end–1) and(a, left_end, k)

D (a, n–left_end–1, k–left_end–1) and (a, left_end, k)

Divide and Conquer GATE-CS-2015 (Set 2)

Discuss it

Question 8 Explanation:

See method 4 of http://www.geeksforgeeks.org/kth-smallestlargest-element-unsorted-array/

Question 9

WRONG

Consider the problem of computing min-max in an unsorted array where min and max are minimum and maximum elements of array. Algorithm A1 can compute min-

max in a1 comparisons without divide and conquer. Algorithm A2 can compute min-max in a2 comparisons by scanning the array linearly. What could be the relation between a1 and a2 considering the worst case scenarios?

A a1 < a2

a1 > a2

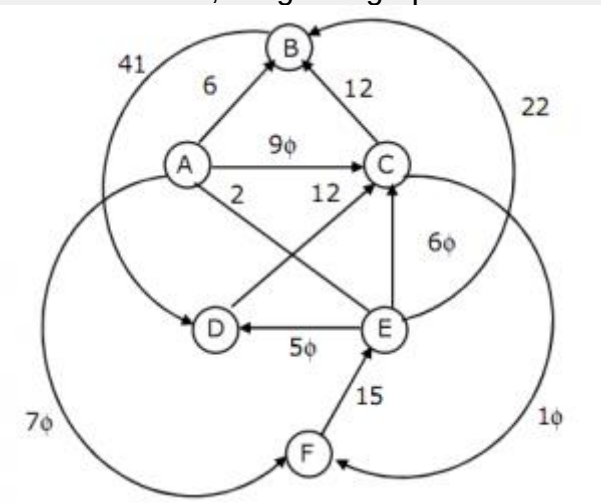
C a1 = a2

Depends on the input

Divide and Conquer GATE 2017 Mock
Discuss it

Question 9 Explanation:
When Divide and Conquer is used to find the minimum-maximum element in an array, Recurrence relation for the number of comparisons is
T(n) = 2T(n/2) + 2 where 2 is for comparing the minimums as well the maximums of the left and right subarrays
On solving, T(n) = 1.5n - 2.
While doing linear scan, it would take 2*(n-1) comparisons in the worst case to find both minimum as well maximum in one pass.

Question 10

Let *G* be the directed, weighted graph shown in below figure 

We are interested in the shortest paths from A. (a) Output the sequence of vertices identified by the Dijkstra's algorithm for single source shortest path when the algorithm is started at node A. (b) Write down sequence of vertices in the shortest path from A to E. (c) What is the cost of the shortest path from A to E?

Divide and Conquer Graph GATE CS 1996
Discuss it

You have completed 9/10 questions .
Your accuracy is 78%.

1

Greedy Algorithms

12

Question 1

CORRECT

Which of the following standard algorithms is not a Greedy algorithm?

A Dijkstra's shortest path algorithm

B Prim's algorithm

C Kruskal algorithm

D Huffman Coding

Bellmen Ford Shortest path algorithm

Greedy Algorithms
Discuss it

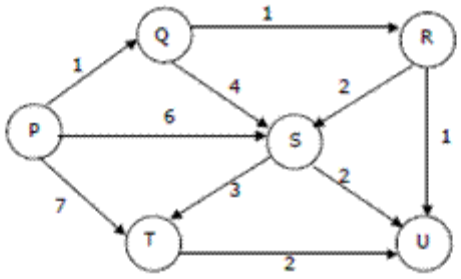
Question 1 Explanation:
See http://www.geeksforgeeks.org/greedy-algorithms-set-1-activity-selection-problem/ and http://www.geeksforgeeks.org/dynamic-programming-set-23-bellman-ford-algorithm/

Question 2

CORRECT

Suppose we run Dijkstra’s single source shortest-path algorithm on the following edge weighted directed graph with vertex P as the source. In what order do the nodes get included

into the set of vertices for which the shortest path distances are finalized? (GATE CS 2004)



A P, Q, R, S, T, U

P, Q, R, U, S, T

C P, Q, R, U, T, S

D P, Q, T, R, U, S

Greedy Algorithms
Discuss it

Question 2 Explanation:
Refer http://www.geeksforgeeks.org/greedy-algorithms-set-6-dijkstras-shortest-path-algorithm/

Question 3

WRONG

A networking company uses a compression technique to encode the message before transmitting over the network. Suppose the message contains the following characters with their frequency:

```
character   Frequency
    a           5
    b           9
    c           12
    d           13
    e           16
    f           45
```

Run on IDE

Note : Each character in input message takes 1 byte. If the compression technique used is Huffman Coding, how many bits will be saved in the message?

A 224

B 800

576

324

Greedy Algorithms
Discuss it

Question 3 Explanation:

```
Total number of characters in the message = 100.

Each character takes 1 byte. So total number of bits needed = 800.


After Huffman Coding, the characters can be represented with:
f: 0
c: 100
d: 101
a: 1100
b: 1101
e: 111
Total number of bits needed = 224
Hence, number of bits saved = 800 - 224 = 576
See here for complete explanation and algorithm.
```

Question 4

WRONG

What is the time complexity of Huffman Coding?

A O(N)

O(NlogN)

O(N(logN)^2)

D O(N^2)

Greedy Algorithms
Discuss it

Question 4 Explanation:
O(nlogn) where n is the number of unique characters. If there are n nodes, extractMin() is called 2*(n – 1) times. extractMin() takes O(logn) time as it calles minHeapify(). So, overall complexity is O(nlogn). See here for more details of the algorithm.

Question 5

WRONG

In question #2, which of the following represents the word *"dead"*?

1011111100101

B 0100000011010

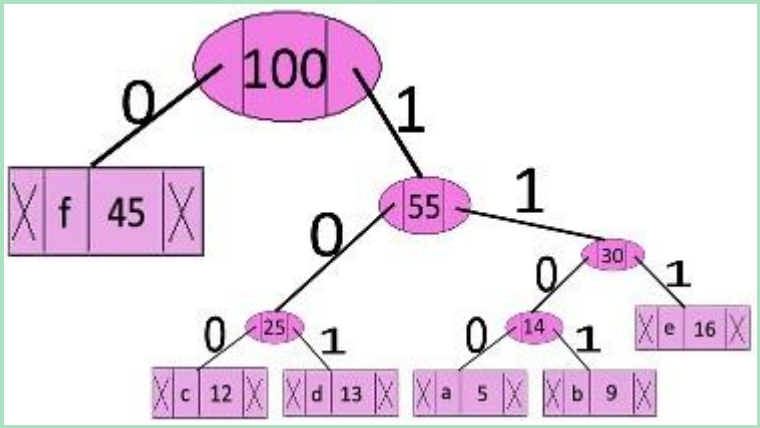
Both A and B

D None of these

Greedy Algorithms
Discuss it

Question 5 Explanation:

The *Huffman Tree* generated is:




```
character   code-word
    f          0
    c          100
    d          101
    a          1100
```

```
b          1101
e          111
```

The word *dead* can be represented as: 101 111 1100 101 However, the alternative codeword can also be found by assigning 1 to the left edge and 0 to the right edge of the tree, i.e. *dead* can also be represented as: 010 000 0011 010 See here for more details of the algorithm.

Question 6

CORRECT

Which of the following is true about Kruskal and Prim MST algorithms? Assume that Prim is implemented for adjacency list representation using Binary Heap and Kruskal is implemented using union by rank.

Worst case time complexity of both algorithms is same.

B Worst case time complexity of Kruskal is better than Prim

C Worst case time complexity of Prim is better than Kruskal

Greedy Algorithms
Discuss it

Question 6 Explanation:
See following Kruskal's Minimum Spanning Tree Algorithm Prim's MST for Adjacency List Representation

Question 7

WRONG

Which of the following is true about Huffman Coding.

A Huffman coding may become lossy in some cases

Huffman Codes may not be optimal lossless codes in some cases

In Huffman coding, no code is prefix of any other code.

D All of the above

Greedy Algorithms
Discuss it

Question 7 Explanation:
Huffman coding is a lossless data compression algorithm. The codes assigned to input characters are Prefix Codes, means the codes are assigned in such a way that the code assigned to one character is not prefix of code assigned to any other character. This is how Huffman Coding makes sure that there is no ambiguity when decoding.

Question 8

CORRECT

Suppose the letters a, b, c, d, e, f have probabilities 1/2, 1/4, 1/8, 1/16, 1/32, 1/32 respectively. Which of the following is the Huffman code for the letter a, b, c, d, e, f?

0, 10, 110, 1110, 11110, 11111

B 11, 10, 011, 010, 001, 000

C 11, 10, 01, 001, 0001, 0000

D 110, 100, 010, 000, 001, 111

Greedy Algorithms GATE-CS-2007
Discuss it

Question 8 Explanation:
We get the following Huffman Tree after applying Huffman Coding Algorithm. The idea is to keep the least probable characters as low as possible by picking them first.

```
The letters a, b, c, d, e, f have probabilities
1/2, 1/4, 1/8, 1/16, 1/32, 1/32 respectively.


                1
              /   \
             /     \
           1/2     a(1/2)
          /  \
         /    \
       1/4   b(1/4)
```

```
        /   \
       /     \
      1/8   c(1/8)
     /  \
    /    \
  1/16  d(1/16)
 /  \
e    f
```

Question 9

WRONG

Suppose the letters a, b, c, d, e, f have probabilities 1/2, 1/4, 1/8, 1/16, 1/32, 1/32 respectively. What is the average length of Huffman codes?

3

B 2.1875

C 2.25

1.9375

Greedy Algorithms GATE-CS-2007

Discuss it

Question 9 Explanation:
We get the following Huffman Tree after applying Huffman Coding Algorithm. The idea is to keep the least probable characters as low as possible by picking them first.

```
The letters a, b, c, d, e, f have probabilities
1/2, 1/4, 1/8, 1/16, 1/32, 1/32 respectively.

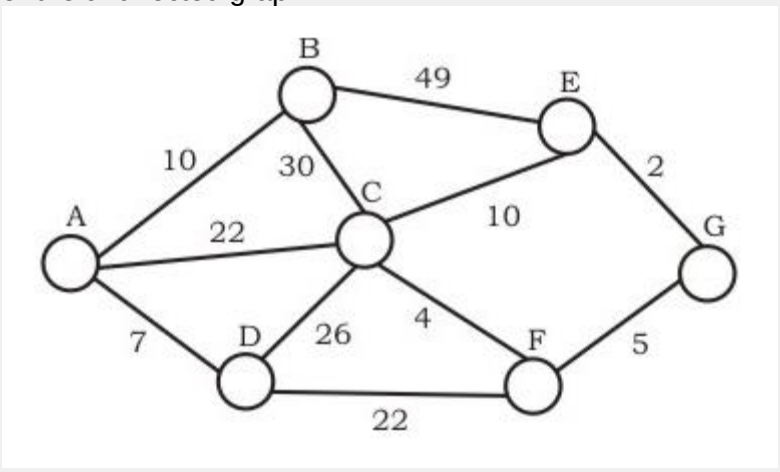

              1
             /   \
            /     \
           1/2    a(1/2)
          /  \
```

```
          /    \
         1/4  b(1/4)
        /   \
       /     \
      1/8   c(1/8)
     /  \
    /    \
  1/16  d(1/16)
 /  \
e    f
The average length = (1*1/2 + 2*1/4 + 3*1/8 + 4*1/16 + 5*1/32 + 5*1/32)
                   = 1.9375
```

Question 10

CORRECT

Consider the undirected graph



below: Using Prim's algorithm to construct a minimum spanning tree starting with node A, which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree?

A (E, G), (C, F), (F, G), (A, D), (A, B), (A, C)

B (A, D), (A, B), (A, C), (C, F), (G, E), (F, G)

C (A, B), (A, D), (D, F), (F, G), (G, E), (F, C)

(A, D), (A, B), (D, F), (F, C), (F, G), (G, E)

Greedy Algorithms GATE-IT-2004
Discuss it

Question 10 Explanation:
A and B are **False** : The idea behind Prim's algorithm is to construct a spanning tree - **means all vertices must be connected** but here vertices are disconnected
C. **False.** Prim's is a **greedy algorithm** and At every step, it considers all the edges that connect the two sets, and picks the minimum weight edge from these edges. In this option weight of AB<AD so must be picked up first. D.**TRUE.** It represents a possible order in which the edges would be added to construct the minimum spanning tree. Prim's algorithm is also a Greedy algorithm. It starts with an empty spanning tree. The idea is to maintain two sets of vertices. The first set contains the vertices already included in the MST, the other set contains the vertices not yet included. At every step, it considers all the edges that connect the two sets, and picks the minimum weight edge from these edges. After picking the edge, it moves the other endpoint of the edge to the set containing MST. Read more at: http://www.geeksforgeeks.org/greedy-algorithms-set-5-prims-minimum-spanning-tree-mst-2/

You have completed 10/15 questions .
Your accuracy is 50%.

12

Greedy Algorithms

12

Question 11

WRONG

Consider the weights and values of items listed below. Note that there is only one unit of each item.

Item number	Weight (in Kgs)	Value (in Rupees)
1	10	60
2	7	28
3	4	20
4	2	24

The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total

value of items picked by an optimal algorithm is denoted by V_{opt}. A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy}. The value of $V_{opt} - V_{greedy}$ is _____ .
Note -This was Numerical Type question.

16

8

C 44

D 60

Greedy Algorithms **Dynamic Programming** **GATE CS 2018**
Discuss it

Question 11 Explanation:

Item No	Weight	Value	Value/Weight
1	10	60	6
2	7	28	4
3	4	20	5
4	2	24	12

After sorting :

Item No	Weight	Value	Value/Weight
4	2	24	12
1	10	60	6
3	4	20	5
2	7	28	4

First we will pick item_4 (Value weight ratio is highest). Second highest is item_1, but cannot be picked because of its weight. Now item_3 shall be picked. item_2 cannot be included because of its weight. Therefore, overall profit by V_{greedy} = 20+24 = 44 Hence, **V_{opt} - V_{greedy} = 60-44 = 16** So, answer is 16.

Question 12

WRONG

A text is made up of the characters a, b, c, d, e each occurring with the probability 0.11, 0.40, 0.16, 0.09 and 0.24 respectively. The optimal Huffman coding technique will have the average length of:

2.40

2.16

C 2.26

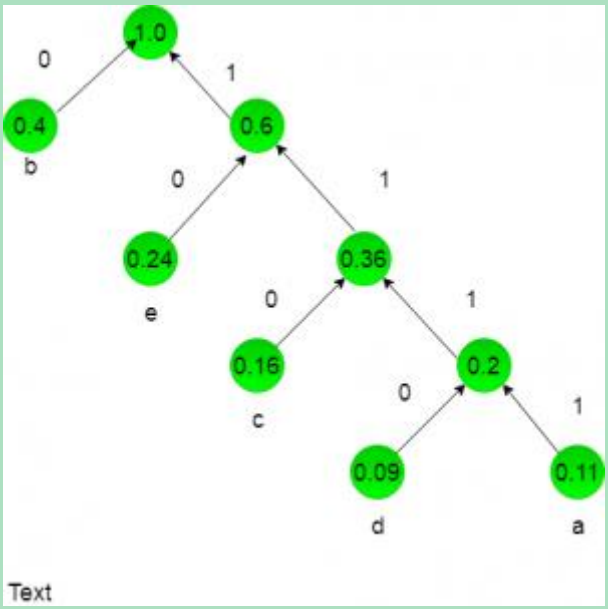
D 2.15

Greedy Algorithms UGC-NET CS 2017 Nov - III

Discuss it

Question 12 Explanation:

a = 0.11 b = 0.40 c = 0.16 d = 0.09 e = 0.24 we will draw a huffman tree:



now huffman coding for character:

```
a = 1111
    b = 0
    c = 110
    d = 1111
    e = 10
lenghth for each character = no of bits * frequency of occurence:
a = 4 * 0.11
  = 0.44
b = 1 * 0.4
```

```
  =  0.4
c = 3 * 0.16
  = 0.48
d = 4 * 0.09
  =  0.36
e = 2 * 0.24
  = 0.48
Now add these lenght for average length:
 0.44 + 0.4 + 0.48 + 0.36 + 0.48 = 2.16
```

So, option (B) is correct.

Question 13

WRONG

Six files F1, F2, F3, F4, F5 and F6 have 100, 200, 50, 80, 120, 150 records respectively. In what order should they be stored so as to optimize act. Assume each file is accessed with the same frequency

F3, F4, F1, F5, F6, F2

B F2, F6, F5, F1, F4, F3

C F1, F2, F3, F4, F5, F6

Ordering is immaterial as all files are accessed with the same frequency.

Greedy Algorithms ISRO CS 2015
Discuss it

Question 13 Explanation:

```
This question is based on the Optimal Storage on Tape problem
which uses greedy approach to find the optimal time to retrieve them.
There are n programs of length L that are to be stored on a computer
tape. Associated with each program i is a length Li. So in order to
retrieve these programs most optimally, we need to store them in the
non-decreasing order of length Li.
So, the correct order is F3, F4, F1, F5, F6, F2
```

Question 14

WRONG

Consider a job scheduling problem with 4 jobs J_1, J_2, J_3, J_4 and with corresponding deadlines: (d_1, d_2, d_3, d_4) = (4, 2, 4, 2). Which of the following is not a feasible schedule without violating any job schedule?

A J_2, J_4, J_1, J_3

J_4, J_1, J_2, J_3

J_4, J_2, J_1, J_3

D J_4, J_2, J_3, J_1

Greedy Algorithms ISRO CS 2007
Discuss it

Question 15

CORRECT

What is the other name of Dijkstra algorithm?

single-source shortest path problem

B multiple-source shortest path problem

C multiple-destination shortest path problem

D single-destination shortest path problem

Greedy Algorithms
Discuss it

You have completed 15/15 questions .
Your accuracy is 40%.

Dynamic Programming

12

Question 1

CORRECT

Which of the following standard algorithms is not Dynamic Programming based.

A Bellman–Ford Algorithm for single source shortest path

B Floyd Warshall Algorithm for all pairs shortest paths

C 0-1 Knapsack problem

Prim's Minimum Spanning Tree

Dynamic Programming
Discuss it

Question 1 Explanation:
Prim's Minimum Spanning Tree is a Greedy Algorithm. All other are dynamic programming based.

Question 2

WRONG

We use dynamic programming approach when

We need an optimal solution

The solution has optimal substructure

C The given problem can be reduced to the 3-SAT problem

D It's faster than Greedy

Dynamic Programming
Discuss it

Question 2 Explanation:
http://www.geeksforgeeks.org/dynamic-programming-set-2-optimal-substructure-property/ Option (D) is incorrect because Greedy algorithms are generally faster than Dynamic programming. See http://www.geeksforgeeks.org/greedy-algorithms-set-1-activity-selection-problem/

Question 3

CORRECT

An algorithm to find the length of the longest monotonically increasing sequence of numbers in an array A[0 :n-1] is given below. Let Li denote the length of the longest monotonically increasing sequence starting at index i in the array.

Initialize $L_{n-1} = 1$

For all i such that $0 \le i \le n-2$

$$L_i = \begin{cases} 1 + L_{i+1} & \text{if } A[i] < A[i+1] \\ 1 & \text{Otherwise} \end{cases}$$

Finally the length of the longest monotonically increasing sequence is $\text{Max}(L_0, L_1, \ldots, L_{n-1})$.

Which of the following statements is TRUE?

The algorithm uses dynamic programming paradigm

B The algorithm has a linear complexity and uses branch and bound paradigm

The algorithm has a non-linear polynomial complexity and uses branch and bound

C paradigm

D The algorithm uses divide and conquer paradigm.

Dynamic Programming
Discuss it

Question 3 Explanation:
Please see http://www.geeksforgeeks.org/dynamic-programming-set-3-longest-increasing-subsequence/

Question 4

WRONG

Kadane algorithm is used to find:

A Maximum sum subsequence in an array

Maximum sum subarray in an array

Maximum product subsequence in an array

D Maximum product subarray in an array

Dynamic Programming
Discuss it

Question 4 Explanation:
Kadane algorithm is used to find the maximum sum subarray in an array. It runs in O(n) time complexity. See details of the algorithm here.

Question 5

WRONG

Four matrices M1, M2, M3 and M4 of dimensions pxq, qxr, rxs and sxt respectively can be multiplied is several ways with different number of total scalar multiplications. For example, when multiplied as ((M1 X M2) X (M3 X M4)), the total number of multiplications is pqr + rst + prt. When multiplied as (((M1 X M2) X M3) X M4), the total number of scalar multiplications is pqr + prs + pst. If p = 10, q = 100, r = 20, s = 5 and t = 80, then the number of scalar multiplications needed is

248000

B 44000

19000

D 25000

Dynamic Programming

Discuss it

Question 5 Explanation:

It is basically matrix chain multiplication problem. We get minimum number of multiplications using ((M1 X (M2 X M3)) X M4). Total number of multiplications = 100x20x5 (for M2 x M3) + 10x100x5 + 10x5x80 = 19000.

Question 6

WRONG

The subset-sum problem is defined as follows. Given a set of n positive integers, S = {a1 ,a2 ,a3 ,…,an} and positive integer W, is there a subset of S whose elements sum to W? A dynamic program for solving this problem uses a 2-dimensional Boolean array X, with n rows and W+1 columns. X[i, j],1 <= i <= n, 0 <= j <= W, is TRUE if and only if there is a subset of {a1 ,a2 ,...,ai} whose elements sum to j. Which of the following is valid for 2 <= i <= n and ai <= j <= W?

A $X[i, j] = X[i - 1, j] \vee X[i, j - ai]$

$X[i, j] = X[i - 1, j] \vee X[i - 1, j - ai]$

$X[i, j] = X[i - 1, j] \wedge X[i, j - ai]$

D $X[i, j] = X[i - 1, j] \wedge X[i - 1, j - ai]$

Dynamic Programming

Discuss it

Question 6 Explanation:

X[I, j] (2 <= i <= n and ai <= j <= W), is true if any of the following is true 1) Sum of weights excluding ai is equal to j, i.e., if X[i-1, j] is true. 2) Sum of weights including ai is equal to j, i.e., if X[i-1, j-ai] is true so that we get (j – ai) + ai as j
See http://www.geeksforgeeks.org/dynamic-programming-subset-sum-problem/ for details.

Question 7

CORRECT

In the above question, which entry of the array X, if TRUE, implies that there is a subset whose elements sum to W?

A X[1, W]

B X[n ,0]

X[n, W]

D X[n -1, n]

Dynamic Programming
Discuss it

Question 7 Explanation:
If we get the entry X[n, W] as true then there is a subset of {a1, a2, .. an} that has sum as W. Reference: http://en.wikipedia.org/wiki/Subset_sum_problem

Question 8

WRONG

A sub-sequence of a given sequence is just the given sequence with some elements (possibly none or all) left out. We are given two sequences X[m] and Y[n] of lengths m and n respectively, with indexes of X and Y starting from 0. We wish to find the length of the longest common sub-sequence(LCS) of X[m] and Y[n] as l(m,n), where an incomplete recursive definition for the function l(i,j) to compute the length of The LCS of X[m] and Y[n] is given below:

```
l(i,j) = 0, if either i=0 or j=0

       = expr1, if i,j > 0 and X[i-1] = Y[j-1]
       = expr2, if i,j > 0 and X[i-1] != Y[j-1]
```

A expr1 ≡ l(i-1, j) + 1

expr1 ≡ l(i, j-1)

expr2 ≡ max(l(i-1, j), l(i, j-1))

D expr2 ≡ max(l(i-1,j-1),l(i,j))

Dynamic Programming GATE-CS-2009
Discuss it

Question 8 Explanation:
In Longest common subsequence problem, there are two cases for X[0..i] and Y[0..j]

```
1) The last characters of two strings match.

   The length of lcs is length of lcs of X[0..i-1] and Y[0..j-1]

2) The last characters don't match.

   The length of lcs is max of following two lcs values

   a) LCS of X[0..i-1] and Y[0..j]

   b) LCS of X[0..i] and Y[0..j-1]
```

Question 9

WRONG

Consider two strings A = "qpqrr" and B = "pqprqrp". Let x be the length of the longest common subsequence (not necessarily contiguous) between A and B and let y be the number of such longest common subsequences between A and B. Then x + 10y = ___.

A 33

23

C 43

34

Dynamic Programming GATE-CS-2014-(Set-2)
Discuss it

Question 9 Explanation:
//The LCS is of length 4. There are 3 LCS of length 4 "qprr", "pqrr" and qpqr A subsequence is a sequence that can be derived from another sequence by selecting zero or more elements from it, without changing the order of the remaining elements. Subsequence need not be contiguous. Since the length of given strings A = “qpqrr” and B = “pqprqrp” are very small, we don’t need to build a 5x7 matrix and solve it using dynamic programming. Rather we can solve it manually just by brute force. We will first check whether there exist a subsequence of length 5 since min_length(A,B) = 5. Since there is no subsequence , we will now check for length 4. “qprr”, “pqrr” and “qpqr” are common in both strings. X = 4 and Y = 3 X + 10Y = 34 This solution is contributed by **Pranjul Ahuja**

Question 10

WRONG

Let A1, A2, A3, and A4 be four matrices of dimensions 10 x 5, 5 x 20, 20 x 10, and 10 x 5, respectively. The minimum number of scalar multiplications required to find the product A1A2A3A4 using the basic matrix multiplication method is

1500

B 2000

500

D 100

Dynamic Programming GATE-CS-2016 (Set 2)

Discuss it

Question 10 Explanation:

We have many ways to do matrix chain multiplication because matrix multiplication is associative. In other words, no matter how we parenthesize the product, the result of the matrix chain multiplication obtained will remain the same. Here we have four matrices A1, A2, A3, and A4, we would have: ((A1A2)A3)A4 = ((A1(A2A3))A4) = (A1A2)(A3A4) = A1((A2A3)A4) = A1(A2(A3A4)). However, the order in which we parenthesize the product affects the number of simple arithmetic operations needed to compute the product, or the efficiency. Here, A1 is a 10 × 5 matrix, A2 is a 5 x 20 matrix, and A3 is a 20 x 10 matrix, and A4 is 10 x 5. If we multiply two matrices A and B of order l x m and m x n respectively,then the number of scalar multiplications in the multiplication of A and B will be lxmxn. Then, The number of scalar multiplications required in the following sequence of matrices will be : A1((A2A3)A4) = (5 x 20 x 10) + (5 x 10 x 5) + (10 x 5 x 5) = 1000 + 250 + 250 = 1500. All other parenthesized options will require number of multiplications more than 1500.

You have completed 10/15 questions .

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Dynamic Programming

12

Question 11

CORRECT

Consider the weights and values of items listed below. Note that there is only one unit of each item.

Item number	Weight (in Kgs)	Value (in Rupees)
1	10	60
2	7	28
3	4	20
4	2	24

The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt}. A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy}. The value of $V_{opt} - V_{greedy}$ is _____ .
Note -This was Numerical Type question.

16

B 8

C 44

D 60

Greedy Algorithms **Dynamic Programming** **GATE CS 2018**
Discuss it

Question 11 Explanation:

Item No	Weight	Value	Value/Weight
1	10	60	6
2	7	28	4
3	4	20	5
4	2	24	12

After sorting :

Item No	Weight	Value	Value/Weight
4	2	24	12
1	10	60	6
3	4	20	5
2	7	28	4

First we will pick item_4 (Value weight ratio is highest). Second highest is item_1, but cannot be picked because of its weight. Now item_3 shall be picked. item_2 cannot be included because of its weight. Therefore, overall profit by V_{greedy} = 20+24 = 44 Hence, **V_{opt} - V_{greedy} = 60-44 = 16** So, answer is 16.

Question 12

WRONG

Consider the following two sequences :

X = < B, C, D, C, A, B, C >, and

Y = < C, A, D, B, C, B >

The length of longest common subsequence of X and Y is :

A 5

B 3

4

2

Dynamic Programming UGC-NET CS 2017 Nov - III

Discuss it

Question 13

WRONG

Consider a sequence F_{00} defined as : $F_{00}(0) = 1$, $F_{00}(1) = 1$ $F_{00}(n) = 10 * F_{00}(n - 1) + 100$ $F_{00}(n - 2)$ for $n \geq 2$ Then what shall be the set of values of the sequence F_{00} ?

(1, 110, 1200)

B (1, 110, 600, 1200)

(1, 2, 55, 110, 600, 1200)

D (1, 55, 110, 600, 1200)

Dynamic Programming UGC NET CS 2017 Jan - II

Discuss it

Question 13 Explanation:

$F_{00}(0) = 1$, $F_{00}(1) = 1$
$F_{00}(n) = 10 * F_{00}(n - 1) + 100$
$F_{00}(2) = 10 * F_{00}(1) + 100$
$= 10 * 1 + 100$
$= 10 + 100$
$= 110$
Similarly:
$F_{00}(3) = 10 * F_{00}(2) + 100$
$= 10 * 110 + 100$
$= 1100 + 100$
$= 1200$
The sequence will be (1, 110, 1200).
So, (A) will be the answer.

Question 14

CORRECT

The following paradigm can be used to find the solution of the problem in minimum time: Given a set of non-negative integer, and a value K, determine if there is a subset of the given set with sum equal to K:

A Divide and Conquer

Dynamic Programming

C Greedy Algorithm

D Branch and Bound

Dynamic Programming ISRO CS 2018
Discuss it

Question 14 Explanation:
Given problem is Subset-sum problem in which a set of non-negative integers, and a value sum is given, to determine if there is a subset of the given set with sum equal to given sum. With recursion technique, time complexity of the above problem is exponential. We can solve the problem in Pseudo-polynomial time using Dynamic programming. Refer: Subset Sum Problem Option (B) is correct

Question 15

WRONG

What happens when a top-down approach of dynamic programming is applied to any problem?

A It increases both, the time complexity and the space complexity

It increases the space complexity and decreases the time complexity.

C It increases the time complexity and decreases the space complexity

It decreases both, the time complexity and the space complexity

Dynamic Programming
Discuss it

Question 15 Explanation:
As the mentioned approach uses the memoization technique it always stores the previously calculated values. Due to this, the time complexity is decreased but the space complexity is increased.

You have completed 15/15 questions .
Your accuracy is 33%.

12

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Which of the following is not a backtracking algorithm?

A Knight tour problem

B N queen problem

Tower of hanoi

M coloring problem

Graph Traversals

123

Question 1

CORRECT

Which of the following algorithms can be used to most efficiently determine the presence of a cycle in a given graph ?

Depth First Search

B Breadth First Search

C Prim's Minimum Spanning Tree Algorithm

D Kruskal' Minimum Spanning Tree Algorithm

Graph Traversals
Discuss it

Question 1 Explanation:
See http://www.geeksforgeeks.org/applications-of-depth-first-search/ and http://www.geeksforgeeks.org/depth-first-traversal-for-a-graph/

Question 2

CORRECT

Traversal of a graph is different from tree because

There can be a loop in graph so we must maintain a visited flag for every vertex

B DFS of a graph uses stack, but inorrder traversal of a tree is recursive

C BFS of a graph uses queue, but a time efficient BFS of a tree is recursive.

D All of the above

Graph Traversals
Discuss it

Question 2 Explanation:
See http://www.geeksforgeeks.org/depth-first-traversal-for-a-graph/

Question 3

WRONG

What are the appropriate data structures for following algorithms?

```
1) Breadth First Search

2) Depth First Search

3) Prim's Minimum Spanning Tree

4) Kruskal' Minimum Spanning Tree
```

```
1) Stack

2) Queue

3) Priority Queue

4) Union Find
```

```
1) Queue

2) Stack

3) Priority Queue

4) Union Find
```

C

```
1) Stack

2) Queue

3) Union Find

4) Priority Queue
```

D

```
1) Priority Queue

2) Queue

3) Stack

4) Union Find
```

Graph Traversals

Discuss it

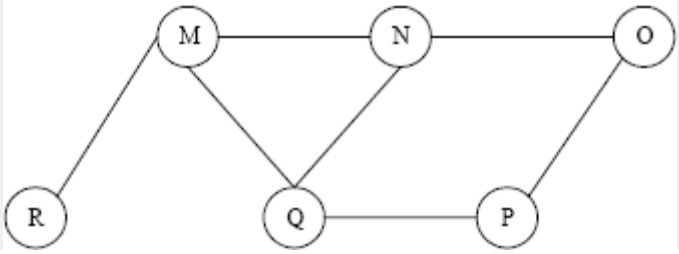
Question 3 Explanation:

1) Breadth First Search uses Queue 2) Depth First Search uses Stack 3) Prim's Minimum Spanning Tree uses Priority Queue. 4) Kruskal' Minimum Spanning Tree uses Union Find.

Question 4

WRONG

The Breadth First Search algorithm has been implemented using the queue data structure. One possible order of visiting the nodes of the following graph is



A MNOPQR

B NQMPOR

QMNPRO

QMNPOR

Graph Traversals
Discuss it

Question 4 Explanation:
Option (A) is MNOPQR. It cannot be a BFS as the traversal starts with M, but O is visited before N and Q. In BFS all adjacent must be visited before adjacent of adjacent. Option (B) is NQMPOR. It also cannot be BFS, because here, P is visited before O. (C) and (D) match up to QMNP. We see that M was added to the queue before N and P (because M comes before NP in QMNP). Because R is M's neighbor, it gets added to the queue before the neighbor of N and P (which is O). Thus, R is visited before O.

Question 5

WRONG

Let G be an undirected graph. Consider a depth-first traversal of G, and let T be the resulting depth-first search tree. Let u be a vertex in G and let v be the first new (unvisited) vertex visited after visiting u in the traversal. Which of the following statements is always true? (GATE CS 2000)

A {u,v} must be an edge in G, and u is a descendant of v in T

{u,v} must be an edge in G, and v is a descendant of u in T

If {u,v} is not an edge in G then u is a leaf in T

D If {u,v} is not an edge in G then u and v must have the same parent in T

Graph Traversals
Discuss it

Question 5 Explanation:

```
In DFS, if 'v' is visited
after 'u', then one of the following is true.
1) (u, v) is an edge.
     u
   /   \
  v     w
 /     / \
x     y   z

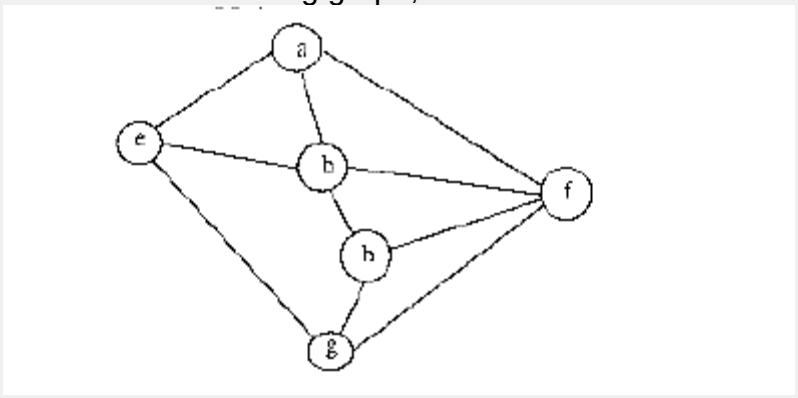
2) 'u' is a leaf node.
     w
   /   \
  x     v
 /     / \
u     y   z
```

In DFS, after visiting a node, we first recur for all unvisited children. If there are no unvisited children (u is leaf), then control goes back to parent and parent then visits next unvisited children.

Question 6

CORRECT

Consider the following graph,



Among the following sequences:

```
(I) a b e g h f

(II) a b f e h g

(III) a b f h g e

(IV) a f g h b e
```

Which are depth first traversals of the above graph?

A I, II and IV only

B I and IV only

C II, III and IV only

I, III and IV only

Graph Traversals
Discuss it

Question 6 Explanation:
We can check all DFSs for following properties.

```
In DFS, if a vertex 'v' is visited
after 'u', then one of the following is true.
1) (u, v) is an edge.
     u
   /   \
  v     w
 /     / \
x     y   z

2) 'u' is a leaf node.
     w
   /   \
  x     v
 /     / \
u     y   z
```

In DFS, after visiting a node, we first recur for all unvisited children. If there are no unvisited children (u is leaf), then control goes back to parent and parent then visits next unvisited children.

Question 7

CORRECT

Make is a utility that automatically builds executable programs and libraries from source code by reading files called makefiles which specify how to derive the target program. Which of the following standard graph algorithms is used by Make.

A Strongly Connected Components

Topological Sorting

C Breadth First Search

D Dijkstra's Shortest Path

Graph Traversals
Discuss it

Question 7 Explanation:
Make can decide the order of building a software using topological sorting. Topological sorting produces the order considering all dependencies provide by makefile. See following for details. Topological Sorting

Question 8

WRONG

Given two vertices in a graph s and t, which of the two traversals (BFS and DFS) can be used to find if there is path from s to t?

A Only BFS

Only DFS

Both BFS and DFS

D Neither BFS nor DFS

Graph Traversals
Discuss it

Question 8 Explanation:
We can use both traversals to find if there is a path from s to t.

Question 9

CORRECT

Which of the following condition is sufficient to detect cycle in a directed graph?

A There is an edge from currently being visited node to an already visited node.

There is an edge from currently being visited node to an ancestor of currently visited

node in DFS forest.

C Every node is seen twice in DFS.

D None of the bove

Graph Traversals
Discuss it

Question 9 Explanation:
See Detect Cycle in a Graph

Question 10

CORRECT

Is following statement true/false If a DFS of a directed graph contains a back edge, any other DFS of the same graph will also contain at least one back edge.
Source: http://courses.csail.mit.edu/6.006/oldquizzes/solutions/q2-s2009-sol.pdf

True

B False

Graph Traversals
Discuss it

Question 10 Explanation:
A back edge means a cycle in graph. So if there is a cycle, all DFS traversals would contain at least one back edge.

Graph Traversals

123
Question 11

WRONG

Is following statement true/false? A DFS of a directed graph always produces the same number of tree edges, i.e., independent of the order in which vertices are considered for DFS.
(Source http://courses.csail.mit.edu/6.006/oldquizzes/solutions/q2-f2008-sol.pdf)

True

False

Graph Traversals
Discuss it

Question 11 Explanation:
Consider the following graph. If we start from 'a', then there is one tree edge. If we start from 'b', then there is no tree edge. a---->b

Question 12

WRONG

If the DFS finishing time f[u] > f[v] for two vertices u and v in a directed graph G, and u and v are in the same DFS tree in the DFS forest, then u is an ancestor of v in the depth first tree. (Source http://courses.csail.mit.edu/6.006/oldquizzes/solutions/q2-f2007-sol.pdf)

True

False

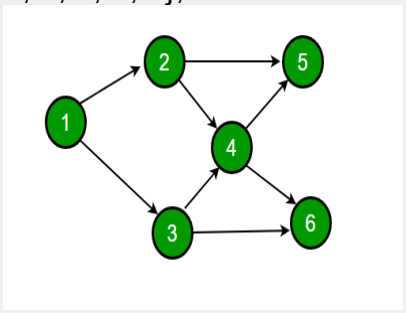
Graph Traversals
Discuss it

Question 12 Explanation:
In a graph with three nodes, r u and v, with edges (r; u) and (r; v), and r is the starting point for the DFS, u and v are siblings in the DFS tree, neither as the ancestor of the other.

Question 13

CORRECT

Consider the DAG with Consider V = {1, 2, 3, 4, 5, 6}, shown below. Which of the following is NOT a topological ordering?



A 1 2 3 4 5 6

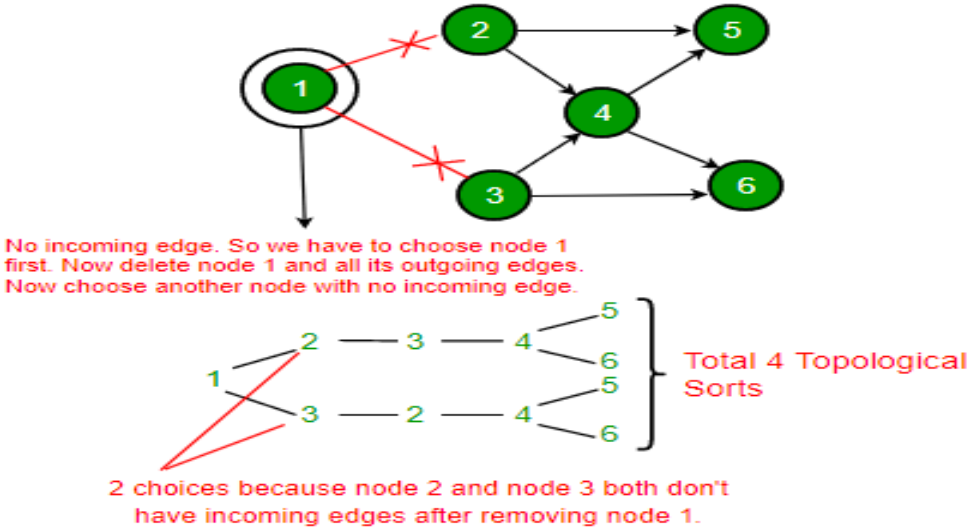
B 1 3 2 4 5 6

C 1 3 2 4 6 5

3 2 4 1 6 5

Graph Traversals GATE-CS-2007
Discuss it

Question 13 Explanation:
In option D, 1 appears after 2 and 3 which is not possible in Topological Sorting. In the given DAG it is directly visible that there is an outgoing edge from vertex 1 to vertex 2 and 3 hence 2 and 3 cannot come before vertex 1 so clearly option D is incorrect topological sort. But for questions in which it is not directly visible we should know how to find topological sort of a
DAG.



This solution is contributed by **Parul sharma.**

Question 14

WRONG

Let G be a graph with n vertices and m edges. What is the tightest upper bound on the running time on Depth First Search of G? Assume that the graph is represented using adjacency matrix.

A $O(n)$

$O(m+n)$

$O(n^2)$

D $O(mn)$

Graph Traversals **GATE-CS-2014-(Set-1)**
Discuss it

Question 14 Explanation:
Depth First Search of a graph takes O(m+n) time when the graph is represented using adjacency list. In adjacency matrix representation, graph is represented as an "n x n" matrix. To do DFS, for every vertex, we traverse the row corresponding to that vertex to find all adjacent vertices (In adjacency list representation we traverse only the adjacent vertices of the vertex). Therefore time complexity becomes $O(n^2)$

Question 15

CORRECT

Consider the tree arcs of a BFS traversal from a source node W in an unweighted, connected, undirected graph. The tree T formed by the tree arcs is a data structure for computing.

A the shortest path between every pair of vertices.

the shortest path from W to every vertex in the graph.

C the shortest paths from W to only those nodes that are leaves of T.

D the longest path in the graph

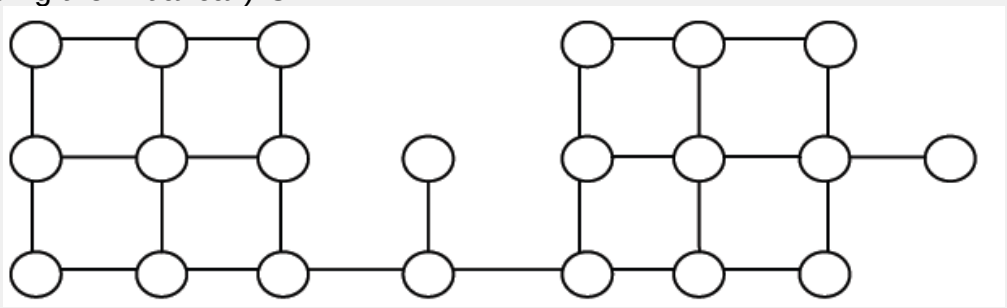
Graph Traversals GATE-CS-2014-(Set-2)
Discuss it

Question 15 Explanation:
BFS always produces shortest path from source to all other vertices in an unweighted graph.

Question 16

WRONG

Suppose depth first search is executed on the graph below starting at some unknown vertex. Assume that a recursive call to visit a vertex is made only after first checking that the vertex has not been visited earlier. Then the maximum possible recursion depth (including the initial call) is



_____.

17

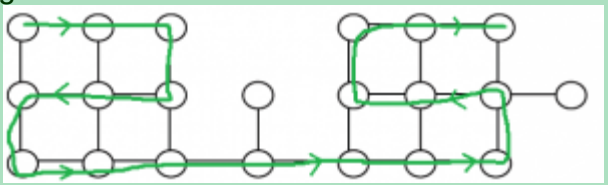
B 18

19

D 20

Graph Traversals GATE-CS-2014-(Set-3)
Discuss it

Question 16 Explanation:

The following diagram shows the worst case situation where the recursion tree has maximum depth.  So the recursion depth is 19 (including the first node).

Question 17

WRONG

Let T be a depth first search tree in an undirected graph G. Vertices u and n are leaves of this tree T. The degrees of both u and n in G are at least 2. which one of the following statements is true?

A There must exist a vertex w adjacent to both u and n in G

B There must exist a vertex w whose removal disconnects u and n in G

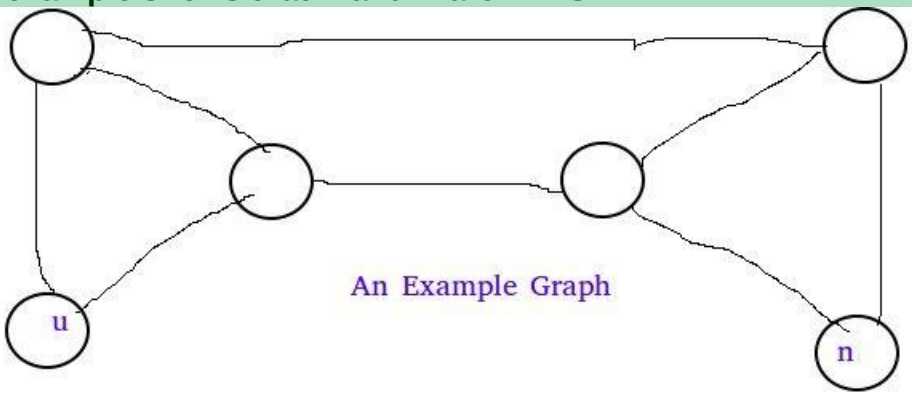
There must exist a cycle in G containing u and n

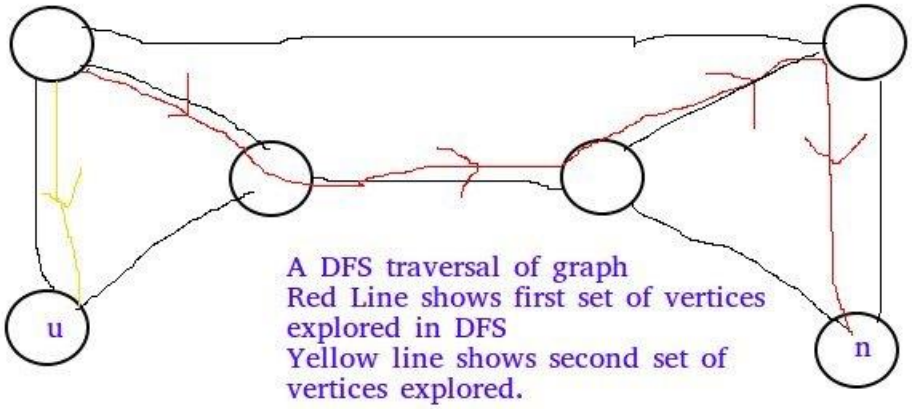
There must exist a cycle in G containing u and all its neighbours in G.

Graph Traversals GATE-CS-2006
Discuss it

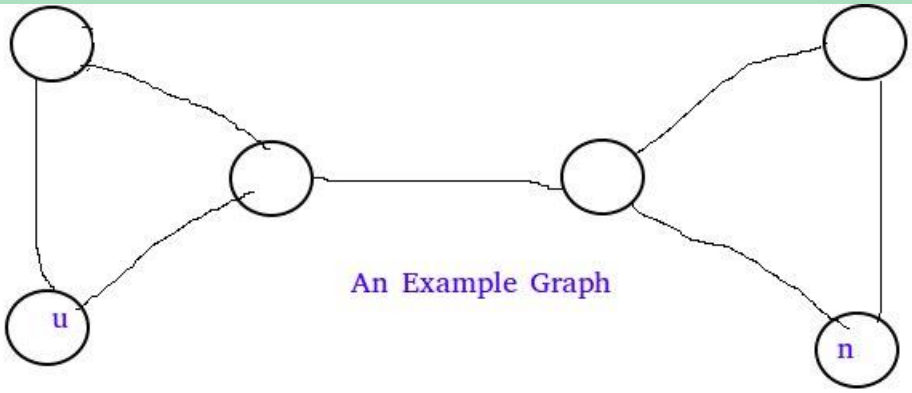
Question 17 Explanation:

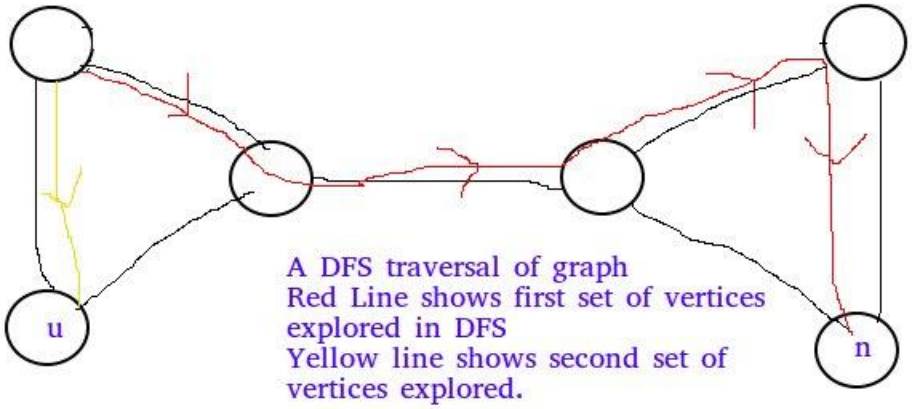
Below example shows that A and B are FALSE:





Below example shows C is false:





Question 18

CORRECT

Let G be an undirected graph. Consider a depth-first traversal of G, and let T be the resulting depth-first search tree. Let u be a vertex in G and let v be the first new (unvisited) vertex visited after visiting u in the traversal. Which of the following statements is always true?

A {u,v} must be an edge in G, and u is a descendant of v in T

B {u,v} must be an edge in G, and v is a descendant of u in T

If {u,v} is not an edge in G then u is a leaf in T

D If {u,v} is not an edge in G then u and v must have the same parent in T

Graph Traversals GATE-CS-2000
Discuss it

Question 18 Explanation:

```
In DFS, if 'v' is visited
after 'u', then one of the following is true.
1) (u, v) is an edge.
     u
   /   \
  v     w
 /     / \
x     y   z

2) 'u' is a leaf node.
     w
   /   \
  x     v
 /     / \
u     y   z
```

In DFS, after visiting a node, we first recur for all unvisited children. If there are no unvisited children (u is leaf), then control goes back to parent and parent then visits next unvisited children.

Question 19

WRONG

In a depth-first traversal of a graph G with n vertices, k edges are marked as tree edges. The number of connected components in G is

k

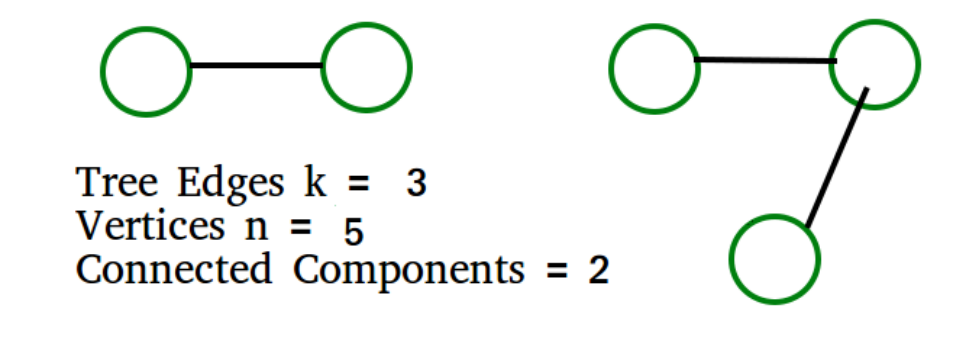
B k + 1

C n - k - 1

n - k

Graph Traversals Gate IT 2005
Discuss it

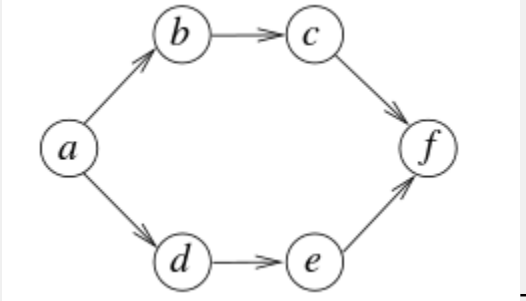
Question 19 Explanation:
Tree edges are the edges that are part of DFS tree. If there are x tree edges in a tree, then x+1 vertices in the tree. The output of DFS is a forest if the graph is disconnected. Let us see below simple example where graph is



disconnected. The above example matches with D option **More Examples:** 1) All vertices of Graph are connected. k must be n-1. We get number of connected components = n- k = n - (n-1) = 1 2) No vertex is connected. k must be 0. We get number of connected components = n- k = n - 0 = n

Question 20

WRONG

Consider the following directed graph. 

The number of different topological orderings of the vertices of the graph is Note : This question was asked as Numerical Answer Type.

1

B 2

C 4

6

Graph Traversals GATE-CS-2016 (Set 1)
Discuss it

Question 20 Explanation:
Following are different 6 Topological Sortings

```
a-b-c-d-e-f

a-d-e-b-c-f

a-b-d-c-e-f

a-d-b-c-e-f

a-b-d-e-c-f

a-d-b-e-c-f
```

You have completed 20/21 questions .

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In an adjacency list representation of an undirected simple graph G = (V, E), each edge (u, v) has two adjacency list entries: [v] in the adjacency list of u, and [u] in the adjacency list of v. These are called twins of each other. A twin pointer is a pointer from an adjacency list entry to its twin. If |E| = m and |V | = n, and the memory size is not a constraint, what is the time complexity of the most efficient algorithm to set the twin pointer in each entry in each adjacency list?

A $\Theta(n^2)$

$\Theta(m+n)$

C $\Theta(m^2)$

D $\Theta(n^4)$

Graph Traversals GATE-CS-2016 (Set 2)
Discuss it

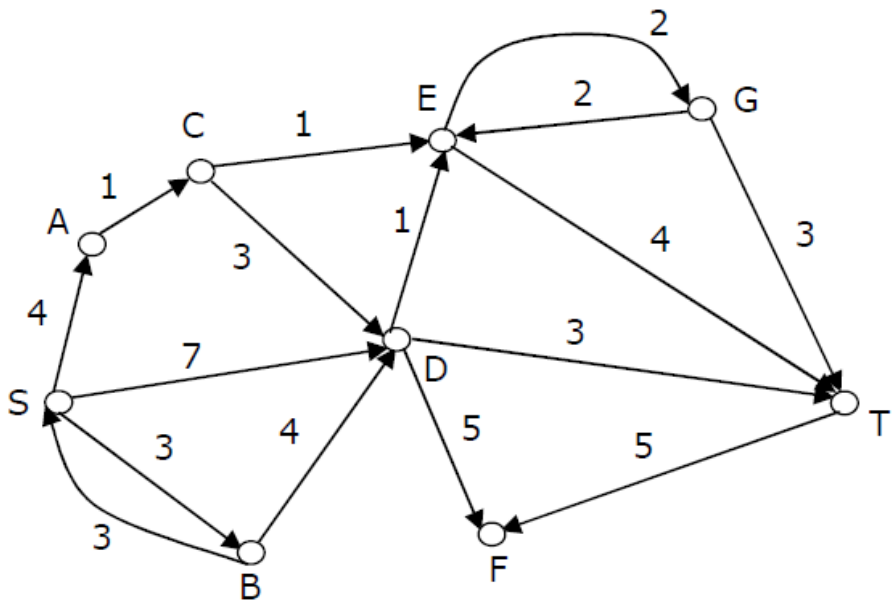
Question 21 Explanation:
First you need to find twins of each node. You can do this using **level order traversal (i.e., BFS)** once. Time complexity of BFS is Θ(m +n). And you have to use linked list for representation which is extra space (but memory size is not a constraint here). Final, time complexity is Θ(m + n) to set twin pointer. Option (B) is correct.

Graph Shortest Paths

Question 1

CORRECT

Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T. Which one will be reported by Dijstra?s shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered.



A SDT

B SBDT

C SACDT

SACET

Graph Shortest Paths
Discuss it

Question 1 Explanation:
See Dijkstra's shortest path algorithm When the algorithm reaches vertex 'C', the distance values of 'D' and 'E' would be 7 and 6 respectively. So the next picked vertex would be 'E'

Question 2

CORRECT

To implement Dijkstra's shortest path algorithm on unweighted graphs so that it runs in linear time, the data structure to be used is:

Queue

B Stack

C Heap

D B-Tree

Graph Shortest Paths

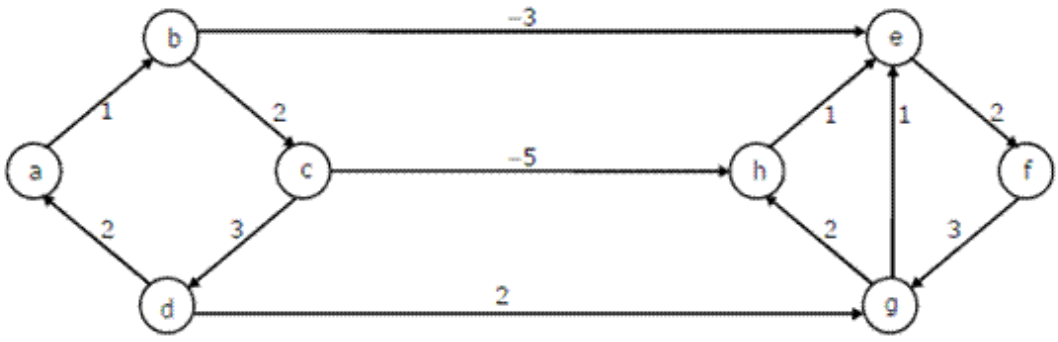
Discuss it

Question 2 Explanation:
The shortest path in an un-weighted graph means the smallest number of edges that must be traversed in order to reach the destination in the graph. This is the same problem as solving the weighted version where all the weights happen to be 1. If we use Queue (FIFO) instead of Priority Queue (Min Heap), we get the shortest path in linear time O(|V| + |E|). Basically we do BFS traversal of the graph to get the shortest paths.

Question 3

CORRECT

Dijkstra's single source shortest path algorithm when run from vertex a in the below graph, computes the correct shortest path distance
to



A only vertex a

B only vertices a, e, f, g, h

C only vertices a, b, c, d

all the vertices

Graph Shortest Paths

Discuss it

Question 3 Explanation:
Dijkstra's single source shortest path is not guaranteed to work for graphs with negative weight edges, but it works for the given graph. Let us see... Let us run the 1st pass b 1 b is minimum, so shortest distance to b is 1. After 1st pass, distances are c 3, e -2. e is minimum, so shortest distance to e is -2 After 2nd pass, distances are c 3, f 0. f is minimum, so shortest distance to f is 0 After 3rd pass, distances are c 3, g 3. Both are same, let us take g. so shortest distance to g is 3. After 4th pass, distances are c 3, h 5 c is minimum, so shortest distance to c is 3 After 5th pass, distances are h -2 h is minimum, so shortest distance to h is -2

Question 4

WRONG

In an unweighted, undirected connected graph, the shortest path from a node S to every other node is computed most efficiently, in terms of time complexity by

Dijkstra's algorithm starting from S.

B Warshall's algorithm

C Performing a DFS starting from S.

Performing a BFS starting from S.

Graph Shortest Paths
Discuss it

Question 4 Explanation:

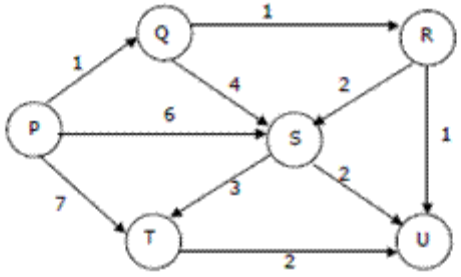
```
 * Time Comlexity of the Dijkstra's algorithm is O(|V|^2 + E)
 * Time Comlexity of the Warshall's algorithm is O(|V|^3)
 * DFS cannot be used for finding shortest paths
 * BFS can be used for unweighted graphs. Time Complexity for BFS is
O(|E| + |V|)
```

Question 5

CORRECT

Suppose we run Dijkstra's single source shortest-path algorithm on the following edge weighted directed graph with vertex P as the source. In what order do the nodes get included into the set of vertices for which the shortest path distances are

finalized? (GATE CS 2004)



A P, Q, R, S, T, U

P, Q, R, U, S, T

C P, Q, R, U, T, S

D P, Q, T, R, U, S

Graph Shortest Paths
Discuss it

Question 5 Explanation:
Refer http://www.geeksforgeeks.org/greedy-algorithms-set-6-dijkstras-shortest-path-algorithm/

Question 6

CORRECT

What is the time complexity of Bellman-Ford single-source shortest path algorithm on a complete graph of n vertices?

A [Tex]\Theta(n^2)[/Tex]

B [Tex]\Theta(n^2 Logn)[/Tex]

[Tex]\Theta(n^3)[/Tex]

D [Tex]\Theta(n^3 Logn)[/Tex]

Graph Shortest Paths
Discuss it

Question 6 Explanation:

Time complexity of Bellman-Ford algorithm is where V is number of vertices and E is number edges (See this). If the graph is complete, the value of E

becomes . So overall time complexity becomes

Question 7

WRONG

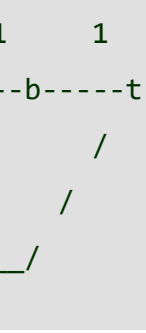
In a weighted graph, assume that the shortest path from a source 's' to a destination 't' is correctly calculated using a shortest path algorithm. Is the following statement true? If we increase weight of every edge by 1, the shortest path always remains same.

Yes

No

Graph Shortest Paths
Discuss it

Question 7 Explanation:
See the following counterexample. There are 4 edges s-a, a-b, b-t and s-t of wights 1, 1, 1 and 4 respectively. The shortest path from s to t is s-a, a-b, b-t. IF we increase weight of every edge by 1, the shortest path changes to s-t.



Question 8

WRONG

Following statement is true or false?

If we make following changes to Dijkstra, then it can be used to find the longest simple path, assume that the graph is acyclic.

1) Initialize all distances as minus infinite instead of plus infinite.

2) Modify the relax condition in Dijkstra's algorithm to update distance
of an adjacent v of the currently considered vertex u only
if "dist[u]+graph[u][v] > dist[v]". In shortest path algo,
the sign is opposite.

True

False

Graph Shortest Paths
Discuss it

Question 8 Explanation:
In shortest path algo, we pick the minimum distance vertex from the set of vertices for which distance is not finalized yet. And we finalize the distance of the minimum distance vertex. For maximum distance problem, we cannot finalize the distance because there can be a longer path through not yet finalized vertices.

Question 9

CORRECT

Which of the following algorithm can be used to **efficiently** calculate single source shortest paths in a Directed Acyclic Graph?

A Dijkstra

B Bellman-Ford

Topological Sort

D Strongly Connected Component

Graph Shortest Paths
Discuss it

Question 9 Explanation:
Using Topological Sort, we can find single source shortest paths in O(V+E) time which is the most efficient algorithm. See following for details. Shortest Path in Directed Acyclic Graph

Question 10

CORRECT

Given a directed graph where weight of every edge is same, we can efficiently find shortest path from a given source to destination using?

Breadth First Traversal

B Dijkstra's Shortest Path Algorithm

C Neither Breadth First Traversal nor Dijkstra's algorithm can be used

D Depth First Search

Graph Shortest Paths
Discuss it

Question 10 Explanation:
With BFS, we first find explore vertices at one edge distance, then all vertices at 2 edge distance, and so on.

You have completed 10/27 questions .

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Graph Shortest Paths

123

Question 11

WRONG

Is the following statement valid about shortest paths? *Given a graph, suppose we have calculated shortest path from a source to all other vertices. If we modify the graph such that weights of all edges is becomes double of the original weight, then the shortest path remains same only the total weight of path changes.*

True

False

Graph Shortest Paths
Discuss it

Question 11 Explanation:
The shortest path remains same. It is like if we change unit of distance from meter to kilo meter, the shortest paths don't change.

Question 12

CORRECT

Match the following

Group A	Group B

```
a) Dijkstra's single shortest path algo                p) Dynamic
Programming

b) Bellmen Ford's single shortest path algo             q) Backtracking

c) Floyd Warshell's all pair shortest path algo.       r) Greedy
Algorithm
```

A a-r, b-q, c-p

B a-p, b-p, c-p

a-r, b-p, c-p

D a-p, b-r, c-q

Graph Shortest Paths
Discuss it

Question 12 Explanation:
Dijkstra is a greedy algorithm where we pick the minimum distant vertex from not yet finalized vertices. Bellman Ford and Floyd Warshell both are Dynamic Programming algorithms where we build the shortest paths in bottom up manner.

Question 13

WRONG

Is the following statement valid?.

Given a weighted graph where weights of all edges are unique (no two edge have same weights), there is always a unique shortest path from a source to destination in such a graph.

True

False

Graph Shortest Paths
Discuss it

Question 13 Explanation:
There can be more than one paths with same weight. Consider a path with one edge of weight 5 and another path with two edges of weights 2 and 3. Both paths have same weights.

Question 14

CORRECT

Is the following statement valid?.

Given a graph where all edges have positive weights, the shortest paths produced by Dijsktra and Bellman Ford algorithm may be different but path weight would always be same.

True

B False

Graph Shortest Paths
Discuss it

Question 14 Explanation:
Dijkstra and Bellman-Ford both work fine for a graph with all positive weights, but they are different algorithms and may pick different edges for shortest paths.

Question 15

WRONG

Which of the following statement(s)is / are correct regarding Bellman-Ford shortest path algorithm?

```
P: Always finds a negative weighted cycle, if one exist s.

Q: Finds whether any negative weighted cycle is reachable

   from the source.
```

P Only

Q Only

C Both P and Q

D Neither P nor Q

Graph Shortest Paths **GATE-CS-2009**
Discuss it

Question 15 Explanation:
Bellman-Ford shortest path algorithm is a single source shortest path algorithm. So it can only find cycles which are reachable from a given source, not any negative weight cycle. Consider a disconnected where a negative weight cycle is not reachable from the source at all. If there is a negative weight cycle, then it will appear in shortest path as the negative weight cycle will always form a shorter path when iterated through the cycle again and again.

Question 16

WRONG

Let G(V, E) an undirected graph with positive edge weights. Dijkstra's single-source shortest path algorithm can be implemented using the binary heap data structure with time complexity:

A O(| V |2)

B O (| E | + | V | log | V |)

O (| V | log | V |)

O ((| E | + | V |) log | V |)

Graph Shortest Paths GATE-CS-2005
Discuss it

Question 16 Explanation:
See Time Complexity Section of http://www.geeksforgeeks.org/greedy-algorithms-set-7-dijkstras-algorithm-for-adjacency-list-representation/

Question 17

WRONG

Let G = (V, E) be an undirected graph with a subgraph G1 = (V1, El). Weights are assigned to edges of G as follows :

$$w(e) = \begin{cases} 0 \text{ if } e \in E_1 \\ 1 \text{ otherwise} \end{cases}$$

A single-source shortest path algorithm is executed on the weighted graph (V, E, w) with an arbitrary vertex v1 of V1 as the source. Which of the following can always be inferred from the path costs computed?

The number of edges in the shortest paths from v1 to all vertices of G

G1 is connected

C V1 forms a clique in G

D G1 is a tree

Graph Shortest Paths GATE-CS-2003
Discuss it

Question 17 Explanation:
When shortest path shortest path from v1 (one of the vertices in V1) is computed. G1 is connected if the distance from v1 to any other vertex in V1 is greater than 0, otherwise G1 is disconnected.

Question 18

WRONG

Let G (V, E) be a directed graph with n vertices. A path from v_i to v_j in G is sequence of vertices (v_i, v_{i+1}, ..., v_j) such that (vk, v_{k+1}) ∈ E for all k in i through j - 1. A simple path is a path in which no vertex appears more than once. Let A be an n x n array initialized as follow

$$A[j,k] = \begin{cases} 1 \text{ if } (j,k) \in E \\ 0 \text{ otherwise} \end{cases}$$

Consider the following algorithm.

```
for i = 1 to n
   for j = 1 to n
      for k = 1 to n
         A [j , k] = max (A[j, k] (A[j, i] + A [i, k]);
```

Which of the following statements is necessarily true for all j and k after terminal of the above algorithm ?

A $A[j, k] \leq n$

B If $A[j, k] \geq n - 1$, then G has a Hamiltonian cycle

If there exists a path from j to k, A[j, k] contains the longest path lens from j to k

If there exists a path from j to k, every simple path from j to k contain most A[j, k] edges

Graph Shortest Paths GATE-CS-2003

Discuss it

Question 18 Explanation:

```
In the original input matrix,  A[j , k] is 1 if there
is an edge from j to k, else 0.


Below expression is important to note:


A[j , k] = max(A[j, k] (A[j, i] + A [i, k]);


This expression puts the count of maximum edges on a path from
j to k.  In this expression, we consider every vertex k that can
become an intermediate vertex and can give longer path.
```

Question 19

WRONG

Let G = (V, E) be a simple undirected graph, and s be a particular vertex in it called the source. For x ∈ V, let d(x) denote the shortest distance in G from s to x. A breadth first search (BFS) is performed starting at s. Let T be the resultant BFS tree. If (u, v) is an edge of G that is not in T, then which one of the following CANNOT be the value of d(u) – d(v)?

-1

B 0

C 1

2

Graph Shortest Paths GATE-CS-2015 (Set 1)
Discuss it

Question 19 Explanation:
Note that the given graph is undirected, so an edge (u, v) also means (v, u) is also an edge. Since a shorter path can always be obtained by using edge (u, v) or (v, u), the difference between d(u) and d(v) can not be more than 1.

Question 20

WRONG

Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j iff either j = i + 1 or j = 3i. The minimum number of edges in a path in G from vertex 1 to vertex 100 is

A 4

7

C 23

99

Graph Shortest Paths Gate IT 2005
Discuss it

Question 20 Explanation:
The task is to find minimum number of edges in a path in G from vertex 1 to vertex 100 such that we can move to either i+1 or 3i from a vertex i.

```
Since the task is to minimize number of edges,
```

```
we would prefer to follow 3*i.

Let us follow multiple of 3.

1 => 3 => 9 => 27 => 81, now we can't follow multiple
of 3. So we will have to follow i+1. This solution gives
a long path.

What if we begin from end, and we reduce by 1 if
the value is not multiple of 3, else we divide by 3.
100 => 99 => 33 => 11 => 10 => 9 => 3 => 1

So we need total 7 edges.
```

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Graph Shortest Paths

123

Question 21

CORRECT

Consider the weighted undirected graph with 4 vertices, where the weight of edge {i, j} g is given by the entry W_{ij} in the matrix W

$$W = \begin{bmatrix} 0 & 2 & 8 & 5 \\ 2 & 0 & 5 & 8 \\ 8 & 5 & 0 & x \\ 5 & 8 & x & 0 \end{bmatrix}$$

The largest possible integer value of x, for which at least one shortest path between some pair of vertices will contain the edge with weight x is _____

Note : This question was asked as Numerical Answer Type.

A 8

12

C 10

D 11

Graph Shortest Paths GATE-CS-2016 (Set 1)

Discuss it

Question 21 Explanation:
Let vertices be 0, 1, 2 and 3. x directly connects 2 to 3. The shortest path (excluding x) from 2 to 3 is of weight 12 (2-1-0-3).

Question 22

CORRECT

The Floyd-Warshall algorithm for all-pair shortest paths computation is based on:

A Greedy paradigm.

B Divide-and-Conquer paradigm.

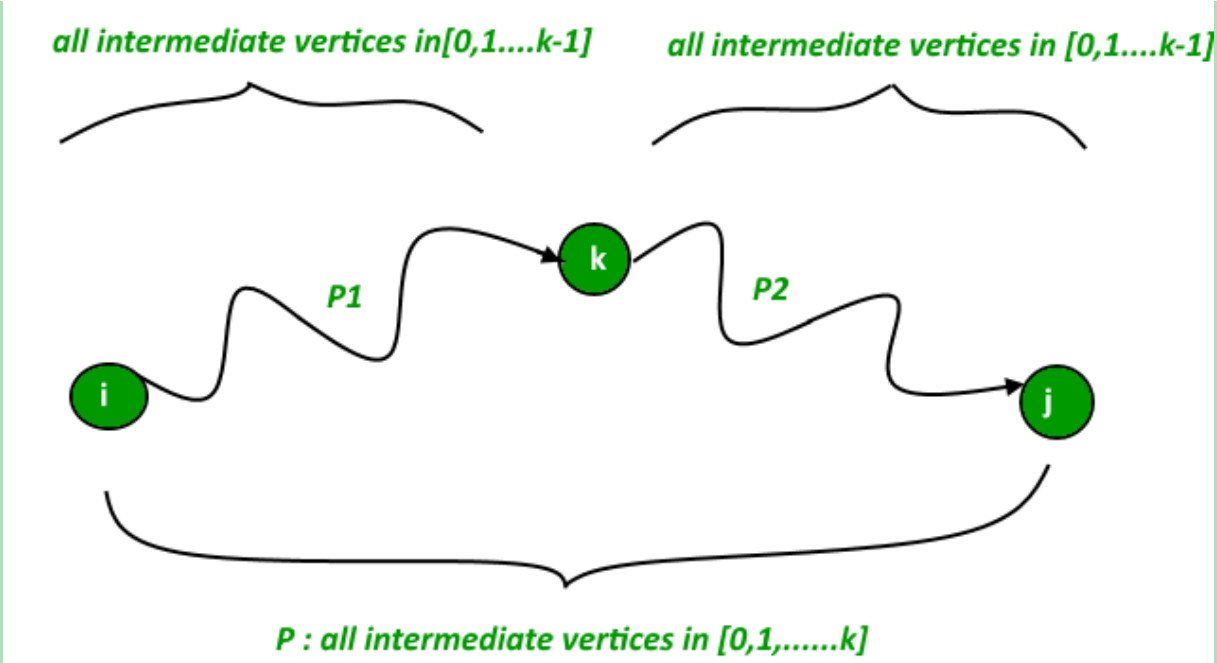
Dynamic Programming paradigm.

D neither Greedy nor Divide-and-Conquer nor Dynamic Programming paradigm.

Graph Shortest Paths GATE-CS-2016 (Set 2)

Discuss it

Question 22 Explanation:
Floyd Warshall Algorithm is a Dynamic Programming based algorithm. It finds all pairs shortest paths using following recursive nature of problem. For every pair (i, j) of source and destination vertices respectively, there are two possible cases. 1) k is not an intermediate vertex in shortest path from i to j. We keep the value of dist[i][j] as it is. 2) k is an intermediate vertex in shortest path from i to j. We update the value of dist[i][j] as dist[i][k] + dist[k][j]. The following figure is taken from the Cormen book. It shows the above optimal substructure property in the all-pairs shortest path problem.



Since there are overlapping subproblems in recursion, it uses dynamic programming.

Question 23

WRONG

Consider a weighted undirected graph with positive edge weights and let uv be an edge in the graph. It is known that the shortest path from the source vertex s to u has weight 53 and the shortest path from s to v has weight 65. Which one of the following statements is always true?

A weight (u, v) < 12

B weight (u, v) ≤ 12

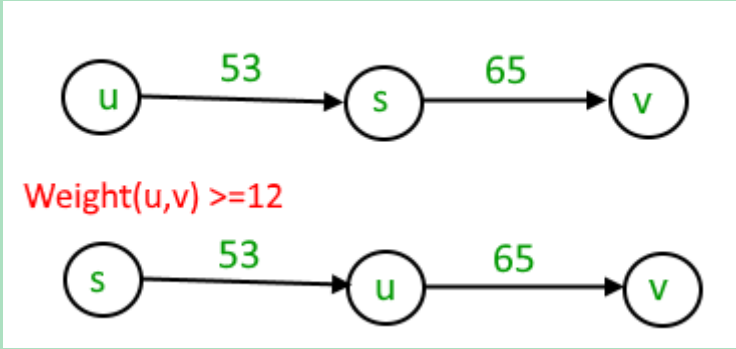
weight (u, v) > 12

weight (u, v) ≥ 12

Graph Shortest Paths Graph Theory Gate IT 2007

Discuss it

Question 23 Explanation:



The minimum weight happens when (S,U) + (U,V) = (S,V) Else (S,U) + (U,V) >= (S,V) Given (S,U) = 53, (S,V) = 65 53 + (U,V) >= 63 (U,V) >= 12. This solution is contributed by **Anil Saikrishna Devarasetty**

Question 24

CORRECT

Which of the following data structure is useful in traversing a given graph by breadth first search?

A Stack

B List

Queue

D None of the above.

Graph Graph Shortest Paths ISRO CS 2017 - May

Discuss it

Question 24 Explanation:
BFS performs level-order traversal which can be fairly done using a queue. A queue uses FIFO ordering and the nodes that we enqueue first are explored first maintaining the order of traversal.

Question 25

CORRECT

Which of the following algorithm solves the all-pair shortest path problem?

A Prim’s algorithm

B Dijikstra’s algorithm

C Bellman-Ford’s algorithm

Floyd-Warshall’s algorithm

Graph Shortest Paths ISRO CS 2017 - May
Discuss it

Question 25 Explanation:
Prim's Algorithm is used to find MST of a given graph. Refer: Prim's Minimum Spanning Tree (MST) Dijikstra's Algorithm is used to find the shortest path from source to all the other nodes in a weighted graph. Refer: Dijkstra's Algorithm BellmanFord's Algorithm is used to find the shortest distance from source to all other nodes in a graph that can contain negetive weight edges. Refer:Bellman–Ford Algorithm Floyd-Warshall's Algorithm: It is an All-pair-shortest path algorithm, used to find shortest distance between every pair of vertices. Refer: Floyd Warshall Algorithm So, option (D) is correct.

Question 26

CORRECT

Which one of the following algorithm design techniques is used in finding all pairs of shortest distances in a graph?

Dynamic programming

B Backtracking

C Greedy

D Divide and Conquer

Graph Shortest Paths ISRO CS 2008
Discuss it

Question 26 Explanation:
Floyd Warshall Algorithm is the All Pairs Shortest Path problem which uses Dynamic Programming to find shortest distances between every pair of vertices in a given edge weighted directed Graph. Option (A) is correct.

Question 27

CORRECT

Which of the following is application of Breath First Search on the graph?

A Finding diameter of the graph

B Finding bipartite graph

Both (a) and (b)

D None of the above

Graph Shortest Paths ISRO CS 2018
Discuss it

Question 27 Explanation:
BFS is used to Find the diameter of the graph and to test whether a graph is bipartite or not. BFS has many other applications also. Refer: Applications of Breadth First Traversal Option (C) is correct.

You have completed 27/27 questions .
Your accuracy is 56%.

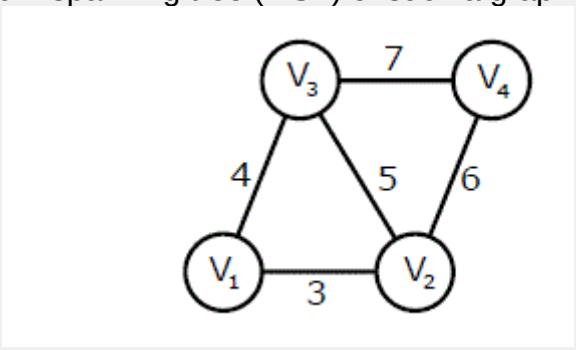
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Graph Minimum Spanning Tree

123
Question 1

WRONG

An undirected graph G(V, E) contains n (n > 2) nodes named v1 , v2 ,….vn. Two nodes vi , vj are connected if and only if 0 < |i – j| <= 2. Each edge (vi, vj) is assigned a weight i + j. A sample graph with n = 4 is shown below. What will be the cost of the minimum spanning tree (MST) of such a graph with n nodes? (GATE CS 2011)



1/12(11n^2 – 5n)

n^2 – n + 1

C 6n – 11

D 2n + 1

Graph Minimum Spanning Tree
Discuss it

Question 1 Explanation:
See http://www.geeksforgeeks.org/data-structures-and-algorithms-set-27/ for explanation.

Question 2

WRONG

The length of the path from v5 to v6 in the MST of previous question with n = 10 is

A 11

25

31

D 41

Graph Minimum Spanning Tree
Discuss it

Question 2 Explanation:
See question 2 of http://www.geeksforgeeks.org/data-structures-and-algorithms-set-27/ for explanation.

Question 3

WRONG

Consider a complete undirected graph with vertex set {0, 1, 2, 3, 4}. Entry Wij in the matrix W below is the weight of the edge {i, j}. What is the minimum possible weight of a spanning tree T in this graph such that vertex 0 is a leaf node in the tree T? (GATE CS 2010)

$$W = \begin{pmatrix} 0 & 1 & 8 & 1 & 4 \\ 1 & 0 & 12 & 4 & 9 \\ 8 & 12 & 0 & 7 & 3 \\ 1 & 4 & 7 & 0 & 2 \\ 4 & 9 & 3 & 2 & 0 \end{pmatrix}$$

A 7

8

C 9

10

Graph Minimum Spanning Tree
Discuss it

Question 3 Explanation:
To get the minimum spanning tree with vertex 0 as leaf, first remove 0th row and 0th column and then get the minimum spanning tree (MST) of the remaining graph. Once we have MST of the remaining graph, connect the MST to vertex 0 with the edge with minimum weight (we have two options as there are two 1s in 0th row).

Question 4

WRONG

In the graph given in above question question, what is the minimum possible weight of a path P from vertex 1 to vertex 2 in this graph such that P contains at most 3 edges?

A 7

8

9

D 10

Graph Minimum Spanning Tree
Discuss it

Question 4 Explanation:
Path: 1 -> 0 -> 4 -> 2 Weight: 1 + 4 + 3

Question 5

CORRECT

An undirected graph G has n nodes. Its adjacency matrix is given by an n × n square matrix whose (i) diagonal elements are 0‘s and (ii) non-diagonal elements are 1‘s. which one of the following is TRUE?

A Graph G has no minimum spanning tree (MST)

B Graph G has a unique MST of cost n-1

Graph G has multiple distinct MSTs, each of cost n-1

D Graph G has multiple spanning trees of different costs

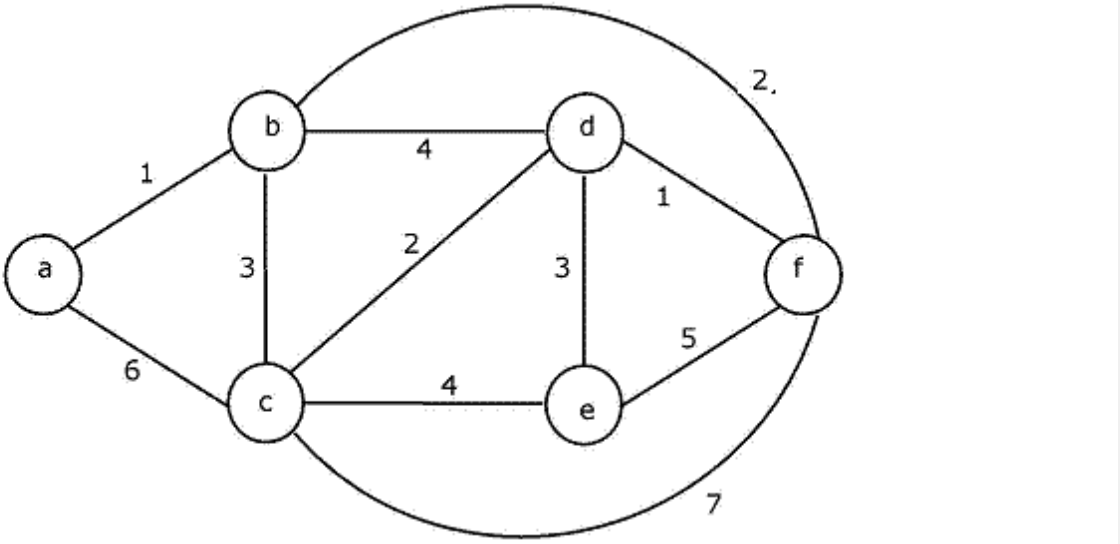
Graph Minimum Spanning Tree
Discuss it

Question 5 Explanation:
See Question 2 of http://www.geeksforgeeks.org/data-structures-and-algorithms-set-22/

Question 6

WRONG

Consider the following graph:



Which one of the following cannot be the sequence of edges added, in that order, to a minimum spanning tree using Kruskal's algorithm?

A (a—b),(d—f),(b—f),(d—c),(d—e)

B (a—b),(d—f),(d—c),(b—f),(d—e)

(d—f),(a—b),(d—c),(b—f),(d—e)

(d—f),(a—b),(b—f),(d—e),(d—c)

Graph Minimum Spanning Tree
Discuss it

Question 6 Explanation:
The edge (d-e) cannot be considered before (d-c) in Kruskal's minimum spanning tree algorithm because Kruskal's algorithm picks the edge with minimum weight from the current set of edges at each step.

Question 7

CORRECT

Let G be an undirected connected graph with distinct edge weight. Let emax be the edge with maximum weight and emin the edge with minimum weight. Which of the following statements is false? (GATE CS 2000)

A Every minimum spanning tree of G must contain emin

B If emax is in a minimum spanning tree, then its removal must disconnect G

No minimum spanning tree contains emax

D G has a unique minimum spanning tree

Graph Minimum Spanning Tree
Discuss it

Question 7 Explanation:
(a) and (b) are always true. (c) is false because (b) is true. (d) is true because all edge weights are distinct for G.

Question 8

WRONG

Consider a weighted complete graph G on the vertex set {v1,v2 ,v} such that the weight of the edge (v,,v) is 2|i-j|. The weight of a minimum spanning tree of G is: (GATE CS 2006)

A n — 1

2n — 2

nC2

D 2

Graph Minimum Spanning Tree
Discuss it

Question 8 Explanation:
Minimum spanning tree of such a graph is

```
v1
  \
    v2
      \
        v3
```

```
          \
           v4
             .
               .
                 .
                   vn
```

Weight of the minimum spanning tree = 2|2 - 1| + 2|3 - 2| + 2|4 - 3| + 2|5 - 4| .... + 2| n - (n-1) | = 2n - 2

Question 9

WRONG

Let G be a weighted graph with edge weights greater than one and G'be the graph constructed by squaring the weights of edges in G. Let T and T' be the minimum spanning trees of G and G', respectively, with total weights t and t'. Which of the following statements is TRUE?

A T' = T with total weight t' = t^2

B T' = T with total weight t' < t^2

T' != T but total weight t' = t^2

None of the above

GATE CS 2012 Graph Minimum Spanning Tree
Discuss it

Question 9 Explanation:
Squaring the weights of the edges in a weighted graph will not change the minimum spanning tree. Assume the opposite to obtain a contradiction. If the minimum spanning tree changes then at least one edge from the old graph G in the old minimum spanning tree T must be replaced by a new edge in tree T' from the graph G' with squared edge weights. The new edge from G' must have a lower weight than the edge from G. This implies that there exists some weights C1 and C2 such that C1 < C2 and C12 >= C22. This is a contradiction.
Source: http://www.cs.nyu.edu/courses/spring06/V22.0310-001/hw3.htm Sums of squares of two or more numbers is always smaller than square of sum. Example: 2^2 + 2^2 < (4)^2 But

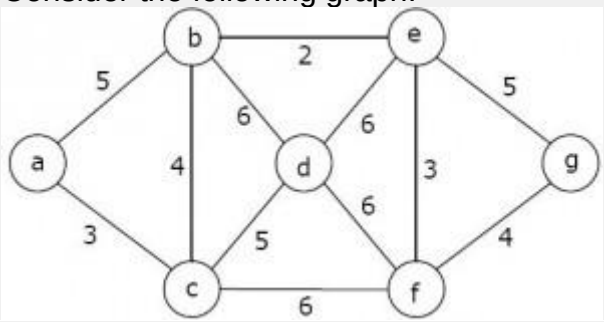
```
there is one counter example when the graph has only one edge.

         In that case, the two values are same.
```

Question 10

CORRECT

Consider the following graph:



Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm?

A (b,e)(e,f)(a,c)(b,c)(f,g)(c,d)

B (b,e)(e,f)(a,c)(f,g)(b,c)(c,d)

C (b,e)(a,c)(e,f)(b,c)(f,g)(c,d)

(b,e)(e,f)(b,c)(a,c)(f,g)(c,d)

Graph Minimum Spanning Tree GATE-CS-2009

Discuss it

Question 10 Explanation:

In the sequence (b, e) (e, f) (b, c) (a, c) (f, g) (c, d) given option D, the edge (a, c) of weight 4 comes after (b, c) of weight 3. In Kruskal's Minimum Spanning Tree Algorithm, we first sort all edges, then consider edges in sorted order, so a higher weight edge cannot come before a lower weight edge.

You have completed 10/21 questions .

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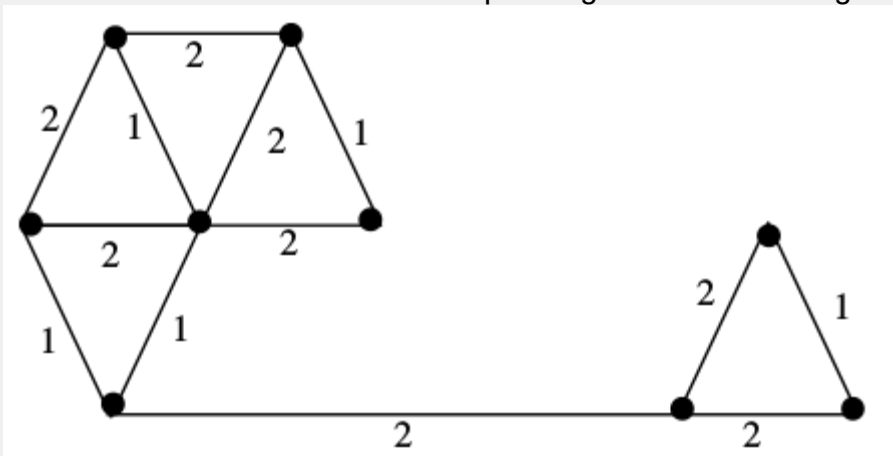
Graph Minimum Spanning Tree

123

Question 11

WRONG

The number of distinct minimum spanning trees for the weighted graph below is ____



4

B 5

6

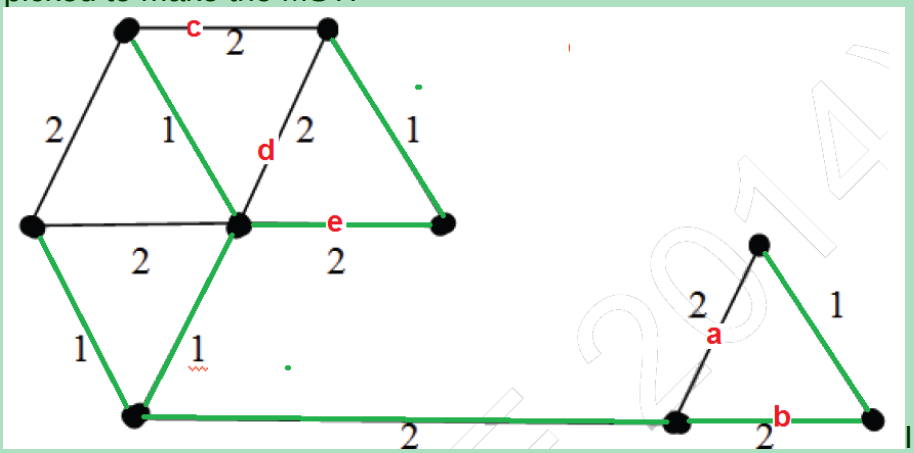
D 7

Graph Minimum Spanning Tree GATE-CS-2014-(Set-2)

Discuss it

Question 11 Explanation:

Below diagram shows a minimum spanning tree. Highlighted (in green) are the edges picked to make the MST.



In the right side of MST, we could either pick edge ‘a’ or ‘b’. In the left side, we could either pick ‘c’ or ‘d’ or ‘e’ in MST. There are 2 options for one edge to be picked and 3 options for another edge to be picked. Therefore, total 2*3 possible MSTs.

Question 12

WRONG

Let s and t be two vertices in a undirected graph G + (V, E) having distinct positive edge weights. Let [X, Y] be a partition of V such that s ∈ X and t ∈ Y. Consider the edge e having the minimum weight amongst all those edges that have one vertex in X and one vertex in Y The edge e must definitely belong to:

the minimum weighted spanning tree of G

B the weighted shortest path from s to t

each path from s to t

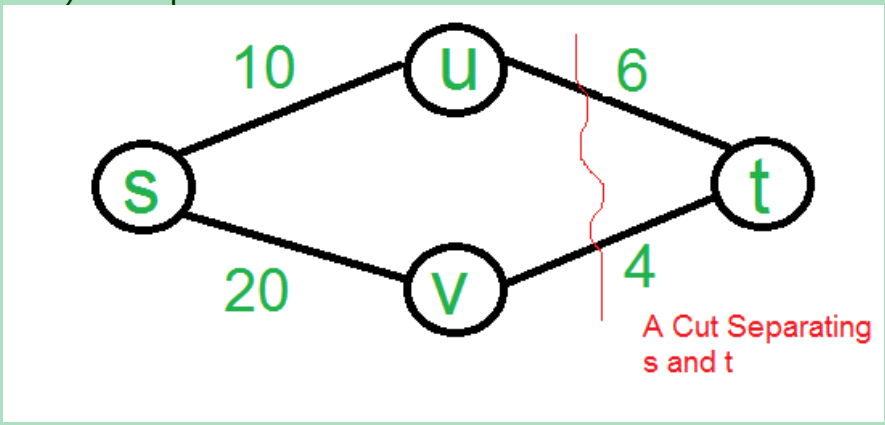
D the weighted longest path from s to t

Graph Minimum Spanning Tree GATE-CS-2005

Discuss it

Question 12 Explanation:

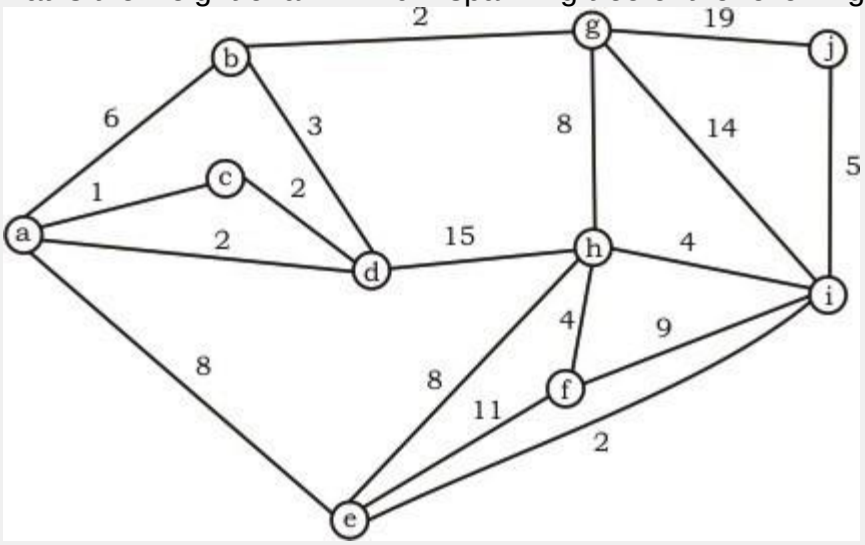
The minimum weight edge on any s-t cut is always part of MST. This is called Cut Property. This is the idea used in Prim's algorithm. The minimum weight cut edge is always a minimum spanning tree edge. **Why B (the weighted shortest path from s to t) is not an answer?** See below example, edge 4 (lightest in highlighted red cut from s to t) is not part of shortest path.



Question 13

WRONG

What is the weight of a minimum spanning tree of the following graph ?



29

31

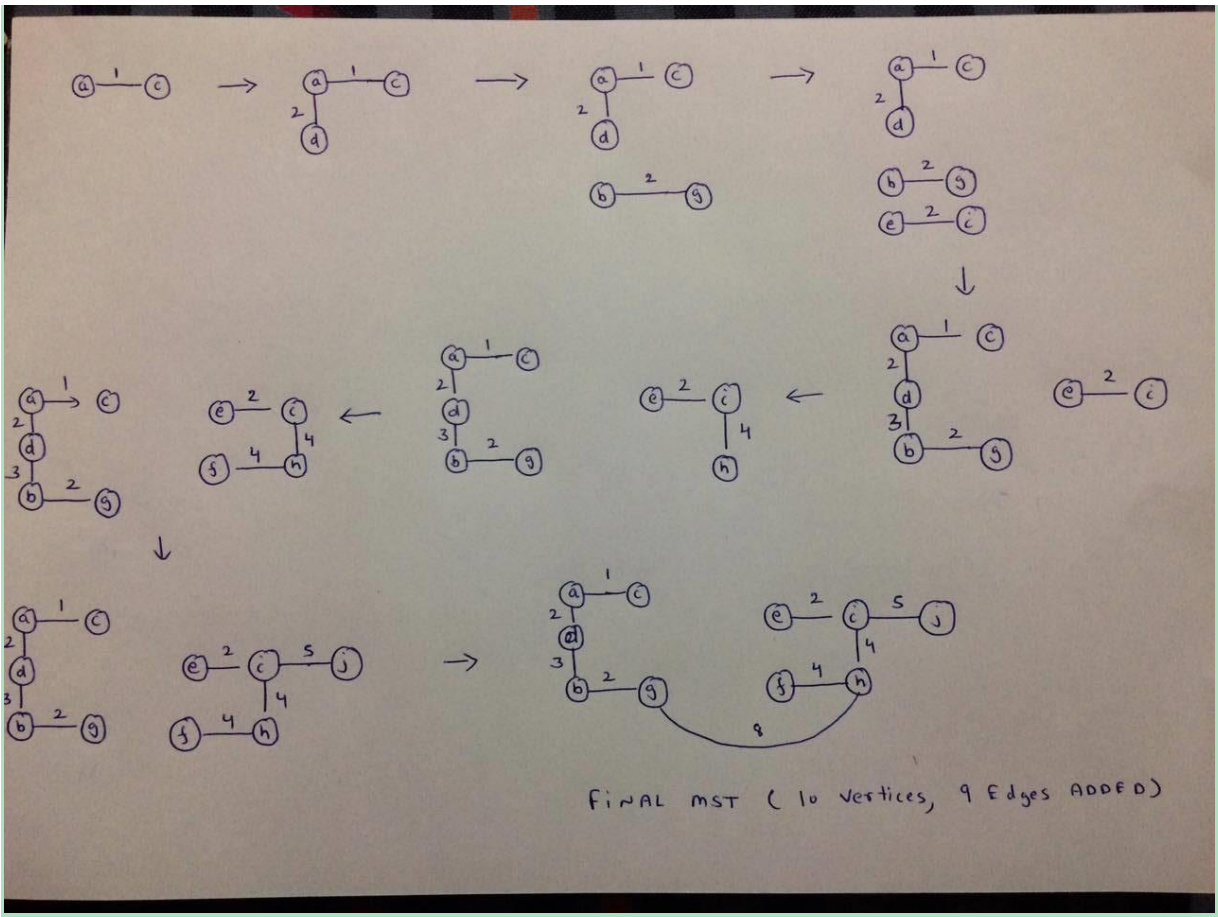
C 38

D 41

Graph Minimum Spanning Tree GATE-CS-2003

Discuss it

Question 13 Explanation:
(a,c), (a,d), (d,b), (b,g), (g,h), (h,f), (h,i), (i,j), (i,e) = 31 Background required - Minimum Spanning Tree (Prims / Kruskal) In these type of questions, always go for kruskal's algorithm to find out the the minimum spanning tree as it is easy and there are less chances of doing silly mistakes. **Algorithm:** Always pick the minimum edge weight and try to add to current forest (Collection of Trees) if no cycle is formed else discard. As soon as u have added n-1 edges to the forest, stop and you have got your minimum spanning tree. See the below image for construction of MST of this question.



Weight of minimum spanning tree = Sum of all the edges in Minimum Spanning tree = 31 This explanation has been provided by **Pranjul Ahuja. Visit the following links to learn more:** http://www.ics.uci.edu/~eppstein/161/960206.html https://en.wikipedia.org/wiki/Minimum_spanning_tree

Question 14

WRONG

Consider an undirected unweighted graph G. Let a breadth-first traversal of G be done starting from a node r. Let d(r,u) and d(r,v) be the lengths of the shortest paths from r to u and v respectively in G. If u is visited before v during the breadth-first traversal, which of the following statements is correct?

d(r, u) < d(r, v)

B d(r,u) > d(r,v)

d(r,u) <= (r,v)

D None of the above

Graph Minimum Spanning Tree **GATE-CS-2001**

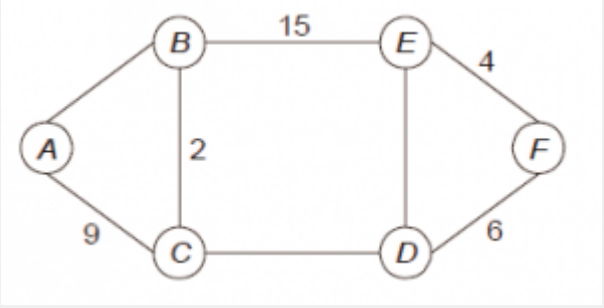
Discuss it

Question 14 Explanation:
See question 4 of http://www.geeksforgeeks.org/data-structures-and-algorithms-set-8/

Question 15

WRONG

The graph shown below 8 edges with distinct integer edge weights. The minimum spanning tree (MST) is of weight 36 and contains the edges: {(A, C), (B, C), (B, E), (E, F), (D, F)}. The edge weights of only those edges which are in the MST are given in the figure shown below. The minimum possible sum of weights of all 8 edges of this graph is _____.



A 66

69

68

D 70

Graph Minimum Spanning Tree GATE-CS-2015 (Set 1)
Discuss it

Question 15 Explanation:
In every cycle, the weight of an edge that is not part of MST must by greater than or equal to weights of other edges which are part of MST. Since all edge weights are distinct, the weight must be greater. So the minimum possible weight of ED is 7, minimum possible weight of CD is 16 and minimum possible weight of AB is 10. Therefore minimum possible sum of weights is 69.

Question 16

WRONG

Let G be connected undirected graph of 100 vertices and 300 edges. The weight of a minimum spanning tree of G is 500. When the weight of each edge of G is increased by five, the weight of a minimum spanning tree becomes _____.

A 1000

995

2000

D 1995

Graph Minimum Spanning Tree GATE-CS-2015 (Set 3)
Discuss it

Question 16 Explanation:
Since there are 100 vertices, there must be 99 edges in Minimum Spanning Tree (MST). When weight of every edge is increased by 5, the increment in weight of MST is = 99 * 5 = 495 So new weight of MST is 500 + 495 which is 995

Question 17

WRONG

Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are TRUE?

```
P: Minimum spanning tree of G does not change

Q: Shortest path between any pair of vertices does not change
```

P only

Q only

C Neither P nor Q

D Both P and Q

Graph Minimum Spanning Tree GATE-CS-2016 (Set 1)
Discuss it

Question 17 Explanation:
The shortest path may change. The reason is, there may be different number of edges in different paths from s to t. For example, let shortest path be of weight 15 and has 5 edges. Let there be another path with 2 edges and total weight 25. The weight of the shortest path is increased by 5*10 and becomes 15 + 50. Weight of the other path is increased by 2*10 and becomes 25 + 20. So the shortest path changes to the other path with weight as 45. The Minimum Spanning Tree doesn't change. Remember the Kruskal's algorithm where we sort the edges first. IF we increase all weights, then order of edges won't change.

Question 18

WRONG

Let G be a complete undirected graph on 4 vertices, having 6 edges with weights being 1, 2, 3, 4, 5, and 6. The maximum possible weight that a minimum weight spanning tree of G can have is. [This Question was originally a Fill-in-the-Blanks question]

6

7

C 8

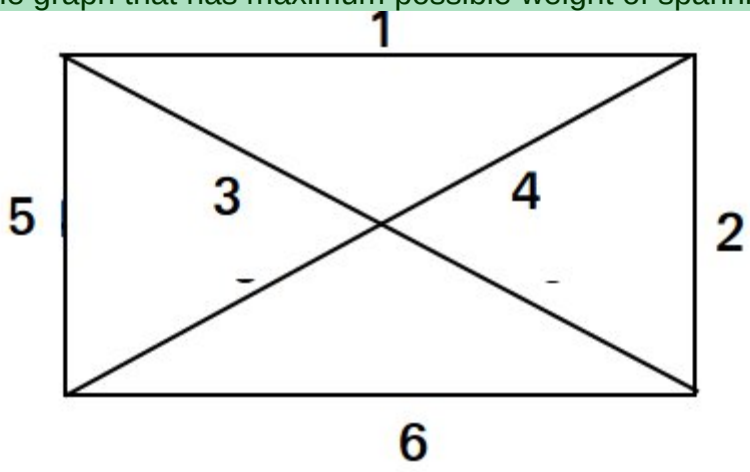
D 9

Graph Minimum Spanning Tree GATE-CS-2016 (Set 1)

Discuss it

Question 18 Explanation:

One graph that has maximum possible weight of spanning tree



Question 19

WRONG

G = (V, E) is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G. Which of the following statements about the minimum spanning trees (MSTs) of G is/are TRUE

```
I.  If e is the lightest edge of some cycle in G,
    then every MST of G includes e
II. If e is the heaviest edge of some cycle in G,
    then every MST of G excludes e
```

A I only

II only

both I and II

D neither I nor II

Graph Minimum Spanning Tree GATE-CS-2016 (Set 1)
Discuss it

Question 19 Explanation:
I is NOT true. Let G=(V, E) be a rectangular graph where V = {a, b, c, d} and E = {ab, bc, cd, da, ac}. Let the edges have weights: ab = 1, bc = 2, cd = 4, da = 5, ac = 3. Then, clearly, ac is the lightest edge of the cycle cdac, however, the MST abcd with cost 7 (= ab + bc + cd) does not include it. Let the edges have weights: ab = 6, bc - 7, cd = 4, da = 5, ac = 3. Then, again, ac is the lightest edge of the cycle cdac, and, the MST bacd with cost 13 (= ba + ac + cd) includes it. So, the MSTs of G may or may not include the lightest edge. **II is true** Let the heavies edge be e. Suppose the minimum spanning tree which contains e. If we add one more edge to the spanning tree we will create a cycle. Suppose we add edge e' to the spanning tree which generated cycle C. We can reduce the cost of the minimum spanning tree if we choose an edge other than e from C for removal which implies that e must not be in minimum spanning tree and we get a contradiction.
Source: http://www.ece.northwestern.edu/~dda902/336/hw5-sol.pdf

Question 20

CORRECT

What is the largest integer m such that every simple connected graph with n vertices and n edges contains at least m different spanning trees?

A 1

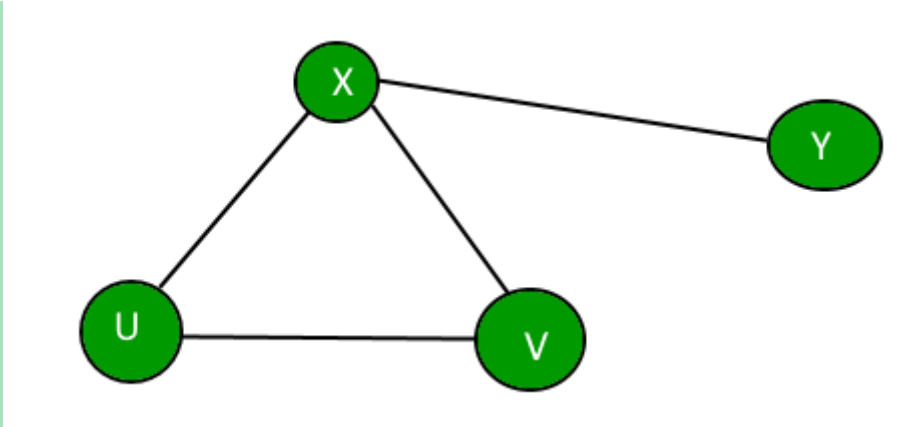
B 2

3

D n

Graph Graph Minimum Spanning Tree Gate IT 2007
Discuss it

Question 20 Explanation:
A graph is connected iff all nodes can be traversed from each node. For a graph with n nodes, there will be n-1 minimum number of edges.
Given that there are n edges, that means a cycle is there in the graph.
The simplex graph with these conditions may be:



Now we can make a different spanning tree by removing one edge from the cycle, one at a time.
Minimum cycle length can be 3, So, there must be atleast 3 spanning trees in any such Graph.

You have completed 20/21 questions .

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If Kruskal's algorithm is used for finding a minimum spanning tree of a weighted graph G with n vertices and m edges and edge weights are already given in a sorted list, then, What will be the time complexity to compute the minimum cost spanning tree given that union and find operations take amortized O(1) ?

A O(m logn)

B O(n)

O(m)

O(n logm)

Graph Minimum Spanning Tree **GATE 2017 Mock**
Discuss it

Question 21 Explanation:
O(m) as you are already given edge weights in sorted order. You just have to pick the edges in the increasing order and add it to the current spanning set if its addition does not result in a cycle else throw it away.